PE7046 Research Project

RETRIEVAL-AUGMENTED GENERATION FOR KNOWLEDGE MANAGEMENT AT LARGE ARCHITECTURE PRACTICES

Module tutor: Dr Ian Watson
Supervisor name: Keerthana Jaganathan
Second marker: Abbas Mehrabidavoodabadi

MSc Computer Science with Artificial Intelligence
Month and year of submission: 12/2025

Student name: **Salvador Cristobal Navarro Perez**
Student ID: **w23051337**

Northumbria University
NEWCASTLE

Retrieval-augmented generation for knowledge management at large architectural practices

Index

Declaration

0 Abstract

1 Introduction

2 Literature review

3 Methodology and prototyping

4 Results and analysis

5 Conclusion and recommendations

+ References and Appendix

DECLARATION

I declare the following:

1. That the material contained in this dissertation is the result of my own work and that due acknowledgement has been given in the bibliography and references to ALL sources be they printed, electronic or personal.

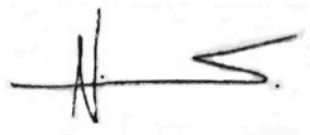
2. I have not use the services of any agency providing model or ghost-written work, nor have I employed unauthorised artificial intelligence in the preparation of this dissertation.

3. The Word Count of this Dissertation is 10,786, excluding figures, tables and appendixes.

4. That unless this dissertation has been confirmed as confidential, I agree to sections of the dissertation being placed on the eLearning Portal (Blackboard), if deemed appropriate, to allow future students the opportunity to see examples of past dissertations. I understand that if displayed on the eLearning Portal it would be made available for no longer than five years and that students would be able to print off copies or download.

5. I agree to my dissertation being submitted to a plagiarism detection service, where it will be stored in a database and compared against work submitted from this or any other Department or from other institutions using the service. In the event of the service detecting a high degree of similarity between content within the service this will be reported back to my supervisor and second supervisor, who may decide to undertake further investigation that may ultimately lead to disciplinary actions, should instances of plagiarism be detected.

6. I have read the Northumbria University Policy Statement on Ethics in Research and I confirm that ethical issues have been considered, evaluated, and appropriately addressed in this research.



SIGNED: Salvador Cristobal Navarro Perez / w23051337
DATE: 17/11/25

0 ABSTRACT

Effective use of Artificial Intelligence (AI) across industries has a potential to increase productivity and shift human resource allocation towards tasks where human employees can shine. Within this broad context, many different implementations of AI are possible. One of these possible applications is Knowledge Management (KM), a strategic area of business management, which focuses on best practices to handle, preserve and transfer intellectual assets within a company, aiming to make it easy for stakeholders to access valuable existing know-how.

One way AI can support KM strategies is through the use of Retrieval-Augmented Generation (RAG), which focuses on natural language communication with a proprietary source of knowledge - a "chat with your data" approach. Although research in this field has existed for years, most literature has focused on studying AI for KM at generic "enterprise environments". By targeting one understudied subset of "enterprise", in this case large Architectural Design (AD) practices, there is an opportunity to study specific needs as well as findings that could generalise to the wider enterprise context.

On this study, a RAG prototype has been developed, evaluated with multiple configurations, and a cost-effective solution that achieves reasonable results for a text-only RAG setup, that relies on good, curated descriptions present on the data, accompanied by a visual user interface (UI), has been presented to domain experts for feedback. The results of all these experiments provide some insight as to what kind of KM tool would be "useful" at a large AD practice and suggest that RAG might, actually, be a viable technology for KM in such context.

1 INTRODUCTION

Artificial Intelligent (AI) powered systems are rapidly getting adoption across all industries. Productivity boost premiums that gain competitive advantage are the holy grail every business dreams of. There are opportunities for AI integration across many existing business processes, and knowledge management (KM) is one of them. The **importance of KM is paramount to knowledge-intensive industries**, such as Architectural Design (AD) - a rather representative subset of the knowledge industry. The case of large architecture practices is particularly relevant, since these are sitting at the i**ntersection of the AD domain and the wider corporate domain** as a whole - they have the specificity of the AD domain, as well as the resources of large corporations - therefore, part of the findings applicable to this context could generalise (with nuances) to both domains.

On that background, an effective KM strategy that makes knowledge easily available to relevant stakeholders within a large architecture practice will save time and resources when new projects revisit proven solutions: when designs encounter situations previously dealt with, when construction systems propose products used on previous projects, when key personnel leaves the company or new starters join in and there is a need for **transfer of know-how**. Development of appropriate KM strategies is still understudied (Yap, Lim and Skitmore, 2022), while it could save significant costs along the design and construction process (Olayinka et al., 2015), by preventing repetition of mistakes, reusing knowledge and **promoting a culture of implementation of lessons learnt**.

For the above reasons, this dissertation studies the application of an *AI-powered KM strategy* that relies on natural language processing for such scenario, in order to answer the research question proposed on 1.1 and find out whether this use case of AI would be worth the effort and necessary resources in the real world.

1.1 Research question

Can a RAG system effectively retrieve and deliver expert-domain information, so that it could be a solution to knowledge management in large architectural practices?

1.2 Aim

Research and select an appropriate RAG architecture, implement it to **evaluate its effectiveness for knowledge management purposes** within the AD domain against a baseline, non-RAG scenario.

1.3 Research objectives

Breakdown of proposed objectives and definition of SMART principles (note the timeline of some of these objectives has overlapped throughout the research life cycle):

1.3.1 To conduct a critical literature review of state-of-the-art RAG techniques

S	Literature review on RAG architectures and knowledge retrieval from PDF documents containing text and image information.
M	A chapter to be part of the research project submission.
A	Literature search using Google Scholar and other relevant search engines. From general to specific.
R	This task helps understand and select suitable architectures for the RAG system.
T	First 16 weeks, plus continous review through the research project life cycle.

1.3.2. To propose success metrics relevant to the research question and define the research method

S	Assessment of valid success metrics to measure research outcome. Definition the appropriate research method.
M	A set of metrics that can be implemented systematically, i.e. within an evaluation suite.
A	Quantifiable metrics for retrieval and generation components (Yu et al., 2024) (Chorev, 2024) (TruEra, 2024) can be tracked. Qualitative test user feedback to be extracted from a survey.
R	By tracking and analysing these metrics and feedback, the effectiveness of the system can be inferred.
T	4 weeks

1.3.3. To collect and process secondary data to feed the RAG

S	Collection of publicly available data about AD projects. Since this research is oriented to enterprise environments, selected projects are related to one particular company.
M	A set of raw PDF documents comprising text and image information.
A	PDFs to be obtained by scraping publicly available websites like archdaily.com or dezeen.com. Alternatively, a publicly available dataset would also be used for benchmarking purposes.
R	Obtained PDF documents would be similar in format and content to those more specific ones inside the company to mimic real-world settings.
T	3 weeks

1.3.4. To shortlist several architecture approaches for the RAG pipeline

S	Based on research, shortlist several setups to be tested and compared. Define their architectures, **including vector database, LLM integration and potential enhancements.**
M	System architecture diagrams showing dependencies between components and identifying the relevant selected technologies for implementation.
A	Achievable through literature review of technical publications.
R	Required for implementation.
T	14 weeks

1.3.5. To build RAG prototypes for evaluation and comparison

S	Build prototypes according to previously shortlisted approaches, including a minimum viable intelligent UI with an AI-powered chatbot feature.
M	Development of the RAG as core part of the research and, additionally, necessary minimum-viable UI elements required for meaningful user interaction with the RAG.
A	Usage of popular, open source libraries and frameworks, local LLM.
R	Essential step to showcase the functionality, measure objective performance of the RAG and conduct evaluation by test users.
T	12 weeks

1.3.6. Evaluation and fine-tuning

S	Evaluate model performance and apply improvements based on initial metric outcome. Extract conclusions at the end.
M	Review of prototype performance based on quantitative objective metrics and qualitative user feedback.
A	Test users would be fellow colleagues in the architecture field who could benefit from such a tool and would be interested in giving valuable feedback on it.
R	Refining the solution would potentially improve the performance (metrics) and user experience, thus enhancing the outcome of the research.
T	12 weeks

1.3.7. To extract conclusions and package the solution for reproducibility

S	To conclude the research with an overall evaluation and a dissemination-ready packaging.
M	Wrapping up the report with final conclusions on the outcome of the research and conveniently packaging the demo.
A	Conclusion and packaging should be just a final step of the process, no blockers foreseen.
R	Necessary step to conclude and come up with the relevant takeaways.
T	4 weeks

1.4 Research approach

Broadly speaking, qualitative, quantitative and mixed methods are the three main research approaches (Creswell, 2014). Given the nature of the problem and proposed solution subject to evaluation (or *artefact* (Hevner et al., 2004)), **this will be a mixed methods approach** attempting to solve a problem from a pragmatic philosophical worldview (Taderhoost, 2022) involving two sequential steps, as follows:

- **Step 1: Quantitative methods** related to objective retrieval and generation performance. Several builds will be compared and a best performer model will be used for the second step (qualitative methods).

- **Step 2: Qualitative methods** on end user experience (i.e. a survey on how users perceive whether the best performing RAG system/tool is helping them, compared to a baseline scenario without such tool).

1.5 Main contributions of this research

The main contributions of this work - aligned with research gaps discussed on 2.7 - are:

1. To contribute to **advance knowledge in cost-effective RAG technology applied to knowledge management, tailored for the AD industry**.
2. To better **understand whether this would be a useful tool for Architects** and **how they would like to interact with it** in real life.

1.6 Limitations of this research

This research has been limited by the below factors:

- Hardware constraints: A small amount of RAG configurations has been tested.
- Granular feature testing, such as dense visual question answering requiring long inference times, have been discarded due to the fact above.
- Data type constraints: Other digital data types, such as BIM and CAD files, information held on digital twins (Xiahou, 2025), MS office files, etc. might be relevant but have been left out of scope due to high technical complexity.
- Some of the experiments, like the exploratory data analysis, have been limited to just the **text corpus** for reasons similar to those discussed above.
- Test user survey has been limited to a small but relevant group of domain experts.

1.7 Structure of this dissertation

The structure from this introduction chapter is as follows: First, a critical literature review section (2), where recent trends and state of the art found on literature have been highlighted, grounds this research. Following that, a methodology chapter (3) proposes how to implement findings and explore research gaps from the literature review. The next chapter (4), shows results obtained from the different experiments. A final section (5) of conclusions and recommendations outlines the main takeaways from this work.

2 LITERATURE REVIEW

Knowledge, in its broadest sense, is a central piece to the success of any organisation. From public institutions to private corporations, “it is widely recognized that **knowledge is an essential strategic resource for a firm to retain sustainable competitive advantage**” (Choi, Poon and Davis, 2008).

Knowledge management (KM) can be defined as “[…] **a discipline that promotes an integrated approach to identifying, capturing, evaluating, retrieving, and sharing all of an enterprise's information assets** […]” (Koenig, 2012). Some subsets of KM are knowledge transfer (i.e. the exchange of knowledge between staff) and knowledge retention (i.e. the ability to keep knowledge after key personnel leave the organisation).

2.1 Current trends in knowledge management and the use of AI

Over time, knowledge is stored on files and formats of all kinds accumulating into, often, fragmented file systems which may lack search assistance capabilities. The exchange of information usually occurs when either “tacit knowledge is communicated through direct person-to person contact” (Choi, Poon and Davis, 2008), or “**explicit-oriented** strategy attempts to increase organizational efficiencies **by codifying and reusing knowledge mainly through advanced ITs**” (Choi, Poon and Davis, 2008).

Suboptimal KM strategies tend to increase costs and man-hours spent on information retrieval tasks - negatively impacting return on investment (ROI) on operations. “Despite the high status of ROI in the business environment […] decision-makers surprisingly avoid evaluating ROI for the most important resource of the organization, notwithstanding the immeasurable hopes that depend on this performance measure and the expectation of establishing the profitability of investment in organizational KM.” (Nakash and Bouhnik, 2022).

Within institutional and enterprise environments, large amounts of domain-specific, valuable data remains inside siloed systems (Wedel, 2025). Recent developments in LLMs are very powerful to provide many kinds of information, but models are usually trained

with, mostly, public domain data, hence they tend to lack the capacity to produce optimal quality answers to domain-specific questions.

Some of the recent trends in knowledge management systems development seek leverage of artificial intelligence (AI) and large language models (LLMs). The “unique capabilities of AI in KM” (Jarrahi et al., 2023) make this combination a high-potential symbiosis, with growing adoption and worth studying in depth.

*An emerging way to bridge this gap is **retrieval-augmented generation** (RAG) technology (Lewis et al., 2020): RAGs can expand the abilities of pre-trained LLMs by harnessing “external and authoritative knowledge bases, rather than solely relying on the model's internal knowledge” (Ng, Matsuba and Zhang, 2025) (Fan et al., 2025).*

Such systems take user queries in natural language, *retrieve* expert-domain content from an external database (*augmenting* the knowledge base of the LLM) and *generate* natural language responses to the queries (Sahin, Varol and Pak, 2024). - “**RAG-based systems draw more on company-specific content than on public internet materials, so there is a lower likelihood of hallucinations**” (Davenport and Bean, 2025). Some researchers have claimed RAG systems, despite their challenges, are one of the most promising technologies for KM, especially in "the domain of FAQs" (Klesel and Whittmann, 2025).

On the other hand, studies also argue that **enterprise RAG implementation presents major challenges** involving "data security, privacy and compliance" (Bruckhaus, 2024), interoperability and domain adaptation (Kumar et al., 2025). Some of these challenges would apply mainly to "compliance-regulated" industries, such as legal or financial services (Bruckhaus, 2024). Another major concern discussed on literature would be a high resource intensity during inference (Khan et al., 2024).

Due to its creative essence, AD services might offer a more forgiving context for their application and success than other industries. The ability to successfully answer questions from retrieved information within processed, embedded mixed-media documents, where

text, images or diagrams might be present, as in real-world documents (Tanaka et al., 2025), would be key to assess the effectiveness for the proposed case scenario.

2.2 RAG systems in KM

As opposed to a fine-tuned LLM approach that would need re-training after certain database updates and would risk overfitting narrow datasets (Khan et al. 2024), **a RAG system would offer greater flexibility of access to external data sources** in real time.

Studies have shown that RAG systems mitigate hallucination rate compared to off-the-shelf LLMs (Dong et al., 2025) and offer greater accuracy in answering domain-specific and time-sensitive enquiries, while also reveal their limitations when "fine-tuning LLMs with small-scale and skewed datasets" (Li, Yuan and Zhang, 2024).

One advantage of RAG against legacy systems is that, while the latter typically lack semantic understanding of searched information, **RAG offers a new paradigm of semantic understanding and user engagement** by setting a conversational context ("chat with your data"). The table below highlights the main differences between the two approaches:

	Traditional KM Systems	**RAG**
Example of tools	Windows Explorer, Open Asset (OA is currently implementing some AI enhancemnets)	Synthesis (Knowledge Architecture, n.d.) targets the architecture industry. However, no information about the inner workings of the software nor performance data is publicly available to the research community
AI integration	None to limited AI or natural language processing (NLP) capabilities	AI-based, integrated NLP
Approach to knowledge management	Not holistic, even fragmented by type of data depending on the tool (i.e. media-only like Open Asset	Holistic (has multi-modality capabilities), potential to intelligently deal with all available data
Similarity / recommendation capabilities	Limited to specific tools. Not available for most basic, widespread tools like Windows Explorer	Great similarity and related information surfacing capabilities

	Traditional KM Systems	RAG
User interface	Folder views or tool-specific	Potentially seamless, conversational and rich-answer capable

Table 2.2 - Comparison between legacy systems and RAG for KM

2.3 Topology of RAG systems: from basic to state-of-the-art

This section introduces the components of a RAG system and looks at basic RAG configurations, moves on to advanced systems that introduce a layer of enhancements, continues to multimodality and, finally, studies research specific to its application in the AD domain.

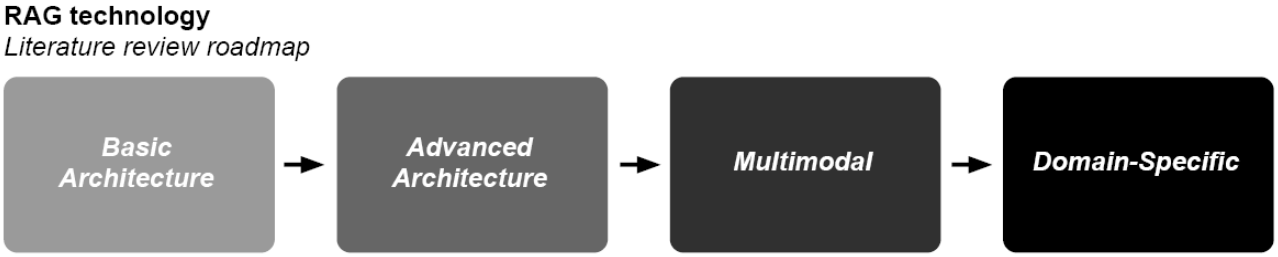


Fig 2.3 - RAG systems review roadmap

2.3.1 Basic architecture of RAG

As a compound AI system (Chen et al., 2025), a basic RAG solution would consist of three essential stages (Bonomo and Bianco, 2025):

A. Pre-Processing (Indexer)

Also referred to as a "pre-retrieval" phase. External knowledge data needs to be appropriately preprocessed - i.e. dividing **documents into chunks** (Bonomo and Bianco, 2025) - and representations of such **data embedded in a vector space** (Mikolov et al., 2013) (Norouzi et al., 2013) (Kiros et al., 2014)). Embeddings stored into a vector database management system (Bonomo and Bianco, 2025) that is computationally efficient (Taipalus, 2024).

B. Retriever

A keystone element of a RAG is the retrieval component (Joshi et al., 2024), which retrieves external knowledge (Chen et al., 2025) from a vector database, actually **locating**

documents by measuring similarity (i.e., using *cosine similarity*). A good quality retrieval will likely lead to a better response. Over recent research, enhancement to retrieval performance has been proposed by, for example, introducing LLM assistance that leads to more adaptive (dynamically adapting to the task at hand) retrieval approaches.

C. Generator

This is **typically a pre-trained transformer model LLM** (Sharma, 2025), which should be able to generate responses over context retrieved from the database (Lewis et al., 2020). This could be a general LLM or a purpose-built, fine-tuned LLM. Examples of pre-trained LLMs (local or accessed via API) include LLaMA, FLAN, T5, GPT-3, GPT-4, Gemini or Claude. Examples of fine-tuned LLMs, which try to improve answer accuracy include ALoFTRAG (Devine, 2025) and Finetune-RAG (Lee, Lin and Tan, 2025).

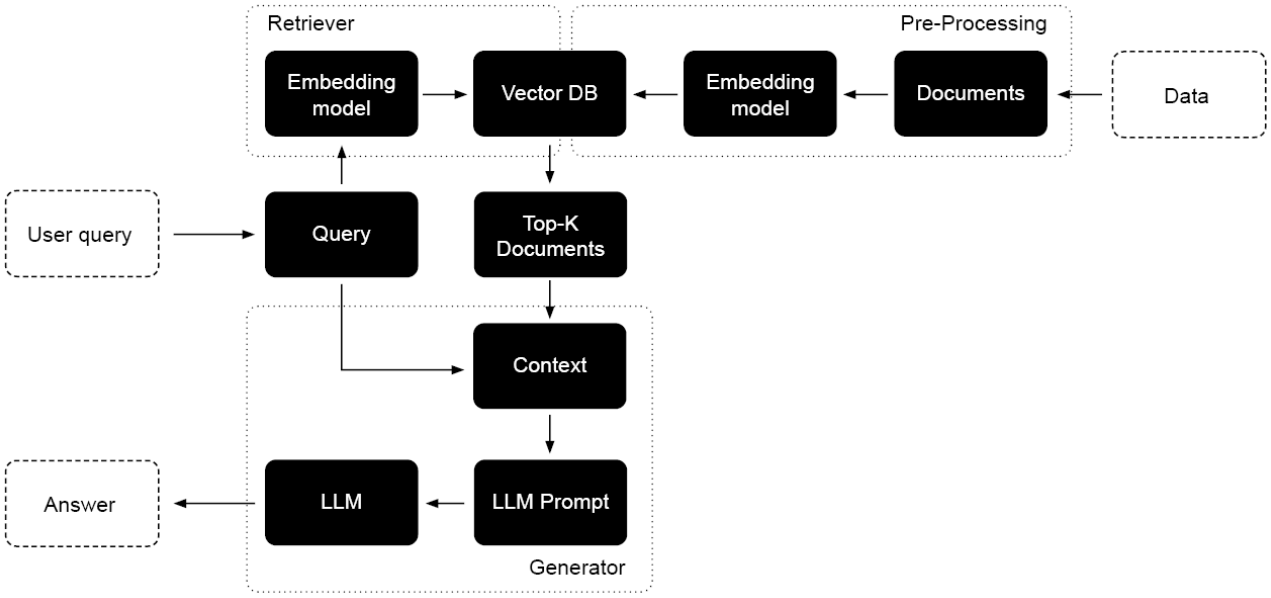


Fig. 2.4.1 - Basic RAG Architecture - Adapted (Hugging Face, n.d.) (IBM, n.d.)

Basic architectures are known to be "struggling with semantic retrieval of the document chunks" (Joshi et al., 2024). Hence, advanced architectures which add further steps to the pipeline have tried to address this problem, although "the influence of various components and configurations within RAG systems remains underexplored" (Li et al, 2025). In other words, **it seems unclear what the specific weight of each component on the overall performance of a RAG system is**.

Studies have argued that universal RAG solutions are not effective for every case and application (Zhao et al., 2025). Instead, these **need to be made specific for the use case** and set of algorithms being used. Depending on the use case, some solutions would even need to be retriever-centric, generator-centric, hybrid or robustness-oriented (Sharma, 2025).

2.3.2 Advanced architecture

Advanced forms of RAG have evolved over time from the basic architecture, aiming to maximise semantic retrieval and reduce hallucinations. Research on semantic retrieval strategies (Zhong et al., 2025) and overall component enhancement along the RAG pipeline has been conducted (Li, Yuan and Zhang, 2024) (Dong et al., 2025). Figure 2.3.2 highlights the components identified on the literature to have the most potential to increase performance:

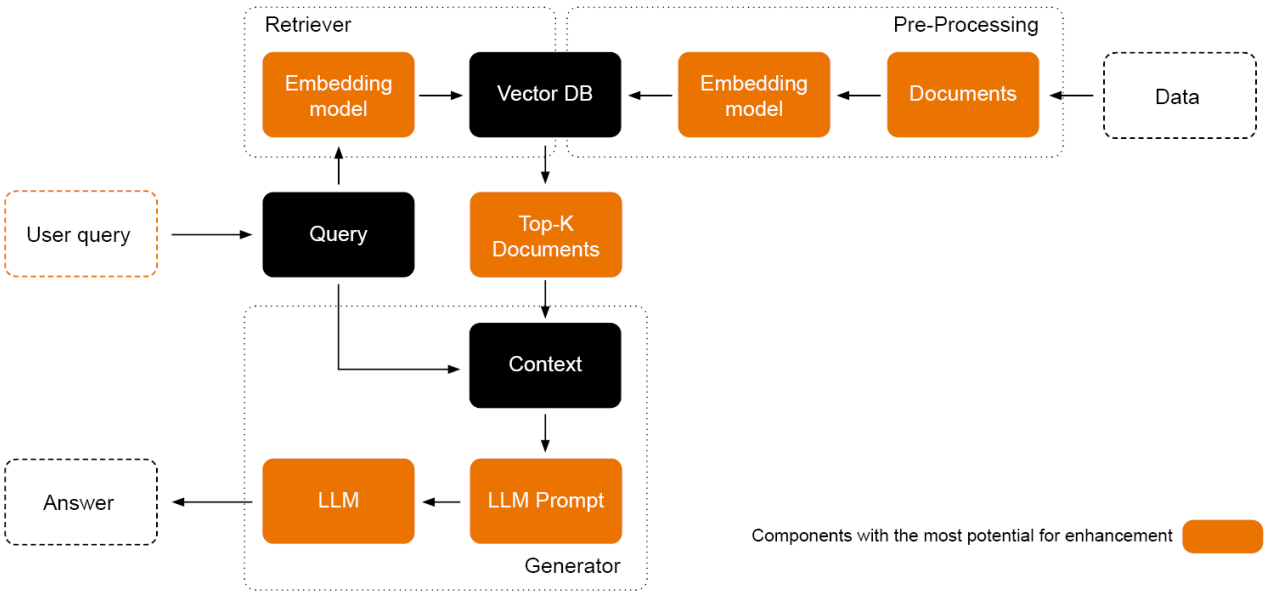


Fig 2.3.2 - Opportunities to enhance a basic RAG architecture

Further review of the impact of different components found on the research is also discussed on the next few paragraphs:

Retrieved volume (the "k" hyperparameter) has a major impact in computational load and overall system performance (Mahapatra and Garain, 2024). Self-reflection has been proposed as an enhancement method that reflects on initially retrieved passages to refine

retrieval (Asai et al., 2024). **Prompt design** is key to task success (Li et al., 2025), indicating that minimal "prompting modifications can yield substantial performance improvements" (Zhao et al., 2025). Prompt enhancement (Chan et al., 2024) and appropriate task framing or few-shot (Brown et al., 2020) strategies and query rewriting (Ma et al., 2023) should be considered in RAG systems.

In regards to **LLM size**, the larger the model, usually the better the responses (Li et al., 2025) (Kaplan et al., 2020). Other studies suggest larger sizes would generate very plausible text but be prone to hallucination and could reduce answer faithfulness on applications other than RAG (Mahapatra and Garain, 2024), although the framed-context nature of RAG prompts could mitigate this issue. Some studies have discussed potential benefits of **contrastive in-context learning**, application of a **focus mode,** or larger **retrieval stride** (Li et al., 2025). In this regard, a two-step retrieval (Pinecone, n.d.) that gets an initially larger set of documents and performs a **reranking** on them, returning a smaller top-k documents might be beneficial.

On the other hand, **document (chunk) size,** sheer **knowledge base size** and **query expansion** do either not have a major impact in performance or provide only marginal gains (Li et al., 2025). Other advanced methods or enhancements added at different stages of the RAG pipeline have been found on literature, such as preprocessing with Differentiable Search Index (DSI) (Tay et al., 2022), the use of a single transformer model to orchestrate indexing and retrieval (Zhuang et al., 2023), advancements in Jaccard/ cosine similarity for vector space retrieval (Jain, et al., 2017), knowledge-augmented language model prompting (KAPING) (Baek, Aji and Saffari, 2023), and task-specific retrievers like Promptagator (Dai et al., 2022).

Main elements prone to enhancement for an "advanced RAG" architecture where some degree of consensus has been found on the research, and might be feasible to apply within the scope of this study, are summarised on table 2.3.2:

Stage	Component	Potential enhancement
Preprocessing	Documents / embedding model	Chunking method settings

	Embedding model	Settings fine-tuning
Retriever	Embedding model	Settings fine-tuning
	Top-K documents	Reranking
Generator	LLM Prompt	Prompt design
	LLM	LLM model choice / size
User query	User query	Query rewriting

Table 2.3.2 - Potential enhancement opportunities along the RAG pipeline

2.3.3 Multimodal architecture

Beyond the advanced architecture for text-based RAGs discussed on 2.3.2, the presence of mixed-media content on the documents requires an additional layer of multimodal capabilities (Joshi et al., 2024). Dealing with various media types has added challenges, compared to dealing with just text sources - such as **enabling caption generation or visual question answering** (VQA) for image information retrieval.

At the intersection of computer vision and natural language processing (NLP), visual question answering (VQA) tasks attempt to answer text-input queries from input images. A successful VQA system typically needs a more complex reasoning system than a system producing only generic captions (Agrawal et al., 2015).

The architecture of a **VQA system typically includes a vision encoder, language encoder and a multimodal fusion machine** (Huynh et al., 2025). CLIP took an approach of "learning directly from raw text about images" (Radford et al., 2021) and represented a breakthrough in multimodality by developing a joint embedding space for image and text. Recent examples of models capable of VQA based on more modern approaches are BLIP (vision-language pre-training) (Li et al., 2022), its upgrade BLIP-2 (Li et al., 2023) (bridges vision-language with LLM language generation) and multimodal large language models (MLLMs) like Flamingo (Alayrac et al., 2022), LLaVA (Li et al., 2022) and InternVL (Kuang et al., 2025).

Multimodal systems then *extract features* from different modalities and appropriate *fusion methods* need to be implemented to produce coherent outputs (Kuang et al., 2025), and context relevance from non-text media that is fed to the LLM is key to the quality of responses (Joshi et al., 2024).

Approaches like MuRAG (Joshi et al., 2024) have attempted to enhance performance of multimodal RAGs (particularly for documents with images tightly interrelated to adjacent text), where page-images with summaries attached are embedded together into a vector database. mKG-RAG (Yuan et al., 2025) is an attempt at providing a robust underlying knowledge structure using **multimodal knowledge graphs** adding a "dual-stage retrieval strategy" that leverages a "coarse-grained document recall" on the wider graph search space with a "fine-grain entity/relationship retrieval".

Visual RAG (Bonomo and Bianco, 2025) is a framework that dynamically expands visual knowledge of multimodal LLMs without requiring fine-tuning, but instead relying on multimodal in-context learning (M-ICL), it uses CLIP to create the image embeddings.

2.3.4 Domain-specific architecture

Some efforts have been made to fine-tune models for the particular field of AD, such as DELF (Noh et al., 2017) and the more recent ArchiCLIP (Xia, Cheng and Tian, 2024), which claims to perform better than general-purpose models for AD feature classification. One of the features of the latter is the use of tags to identify types of media/project/ elements on the images. However, sufficiently capable models that have had a large enough, domain-specific training dataset and are able to successfully perform on long-tail topics within this field seem yet to be developed.

2.3.5 A literature review-based Taxonomy of RAG

Figure 2.3.5 summarises a taxonomy of RAG concepts found on this literature review:

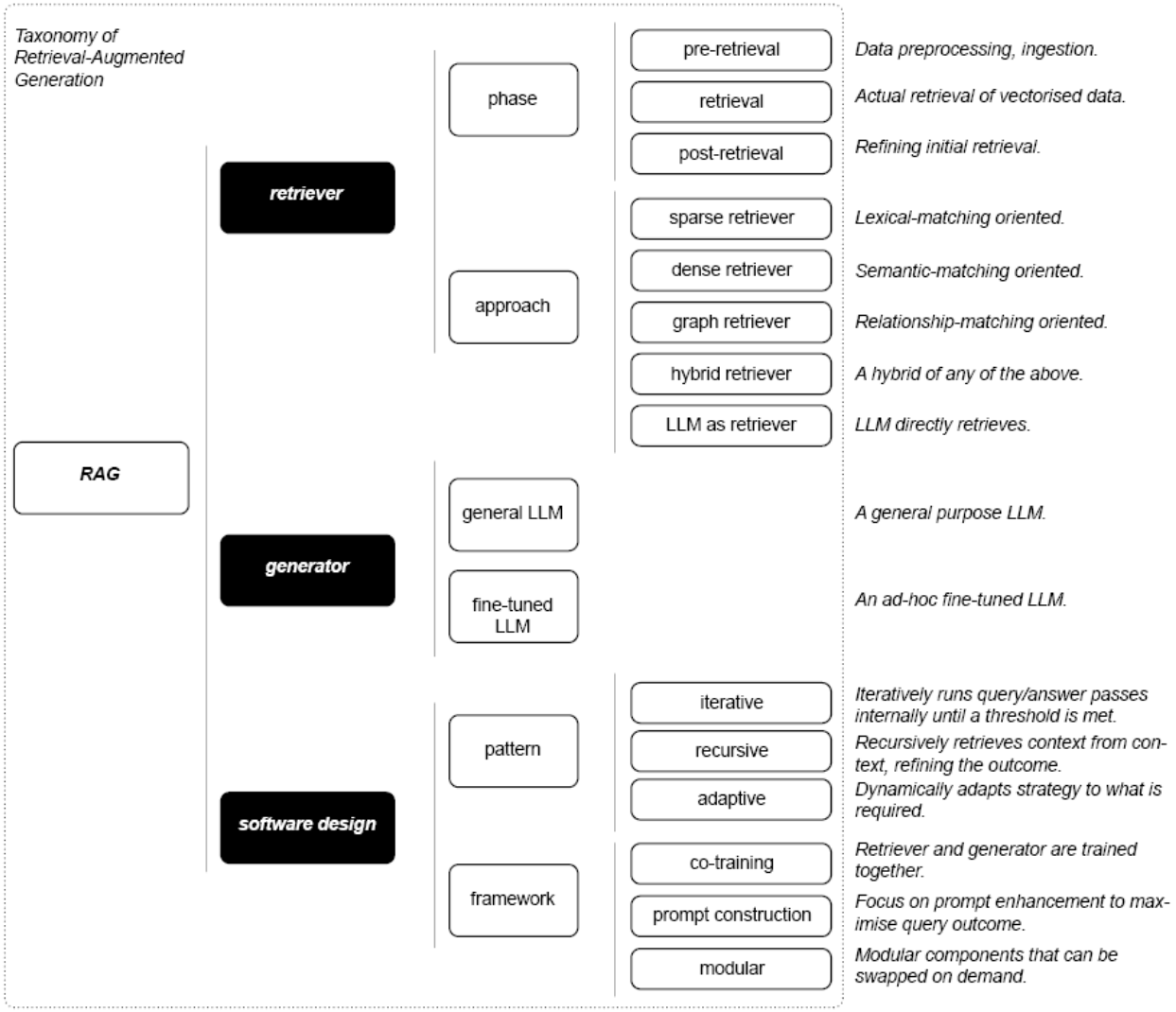


Fig 2.3.5 - Taxonomy of RAG - Adapted from (Chen et al., 2025) (purpleSlate, 2024)

2.4 Measuring success

When measuring success of RAGs, it would also be important to note that, due to the "hybrid structure and reliance on dynamic knowledge sources" (Yu, et al., 2024) of RAGs, it is challenging but equally critical to find appropriate quantifiable metrics.

Studies performing comprehensive grid-search optimisation have shown the "**significance of optimizing the parameters involved in developing RAG-based applications**" (Sakar and Emekci, 2025). Success should be measured on a case-by-case basis, as a weighted balance between performance metrics, run-time efficiency, scalability and hardware

requirements. A set of curated, key performance metrics found on literature relevant to this study has been proposed on chapter 3.8 Evaluation suite.

In terms of qualitative evaluation, recent work can be found that evaluates RAG systems from a user's perspective through human testing (Baur et al., 2025) - albeit results seem promising, this area remains largely under explored.

2.4.1 Benchmarking datasets

RAGBench (Friel, Belyi and Sanyal, 2025) proposes to standardise systematic approaches to **explainable performance evaluation**, with other studies (Joshi et al., 2024) which use **differentiated sets of questions** (S*hort-form-type-QA, Long-form-type-QA, MCQ-type-QA, True-False-type-QA*) propose **recreating different query scenarios**. Other datasets used for benchmarking found on the literature include VisualQA (VQA) (Agrawal et al., 2015) or WixQA (Cohen et al., 2025).

2.5 Limitations of RAG

Some of the limitations and disadvantages of RAG systems are outlined below:

- **Integration between retrieval and generation might sometimes fail**, with generation module not being able to accurately interpret correctly retrieved information (Joshi et al., 2024).
- **Computational cost** increase due to both retrieval and generation steps being computationally expensive (Khan et al., 2024).
- Validation of a RAG system can only be feasible **"during operation"** (Barnett et al., 2024) - rather than design and freeze the system setup from the start.
- **Security vulnerabilities** derived from **database poisoning** (Xue et al., 2024) present on external databases whose content might have been retrieved from the public domain.

2.6 Ethical and legal considerations

Some of the main implications are related to intellectual property protection. From a legal/ ethical point of view, granting access to proprietary company data to a third-party LLM

poses a major risk of data leakage (Bruckhaus, 2024). The use of on-premises, local LLMs mitigates this risk, at the possible cost of loss of accuracy or expensive infrastructure.

2.7 Research gaps found

The main gaps identified during this literature review are highlighted below:

- *Research gap 1:* **RAG application specific to KM for the AD industry is understudied**. In addition, incorporation of standard data formats used in the AD industry like NURBS, mesh, CAD or BIM into RAG systems to RAG is also understudied - however, this area has been left out of this research due to its high diversity and complexity.
- *Research gap 2:* Despite recent advances, like RAGBench (Friel, Belyi and Sanyal, 2025) and user testing trials (Baur et al., 2025), **standardised benchmarking and empirical testing of RAG systems is also understudied**.

3 METHODOLOGY AND PROTOTYPING

In order to assess the feasibility of RAG for KM in AD, a minimum-viable prototype has been built considering findings from the literature review. **The packaged codebase is accessible through this link.** The methodology has followed the below steps:

- Requirement analysis
- System design
- Data collection and processing (data pipeline implementation)
- Exploratory data analysis
- RAG module design
- RAG module implementation
- User interface (UI) implementation (the "App")
- Evaluation suite module implementation
- Test user survey design

3.1 Requirement analysis

The prototype would be designed as a modular software solution, operating an appropriate database, with AI/NLP at its core, so that it would allow users to "chat with their data".

3.1.1. Functional requirements

A. A module to **collect and process** the **data**.

B. A module for **exploratory data analysis** to better understand its contents.

C. A module to **build and populate a SQL database** for the UI from the data.

D. A RAG module central to the prototype, able to retrieve data from its own **vector database** (coordinated with the UI SQL database) and answer questions.

E. A user interface for test users to be able to assess the system.

3.1.2. Performance and evaluation requirements

A. An essential **test suite** to verify correct functionality during development stage.

B. An **evaluation suite** module for optimisation and selection of best performing RAG settings/architecture.

3.2 System design

The system has been designed following usual Python object oriented project development patterns, and code has been divided into appropriate modules, most of which can be run independently. A codebase structure diagram is included on the appendix.

The main programming language selected for this purpose, due to its versatility and AI capabilities, is **Python 3.11.14** (Python Software Foundation, 2025). Other programming languages and web technologies like **JavaScript** (ECMA International, 2025), **HTML** (W3C, 2025) and **CSS** (W3C, 2023) have been used to a lesser extent to support the development of the UI. Extensive refactoring has been conducted to optimise performance, maintainability and robustness of the system throughout the development lifecycle.

Figure 3.2 and table 3.2 show a proposal for a system design consistent with the requirements discussed on 3.1:

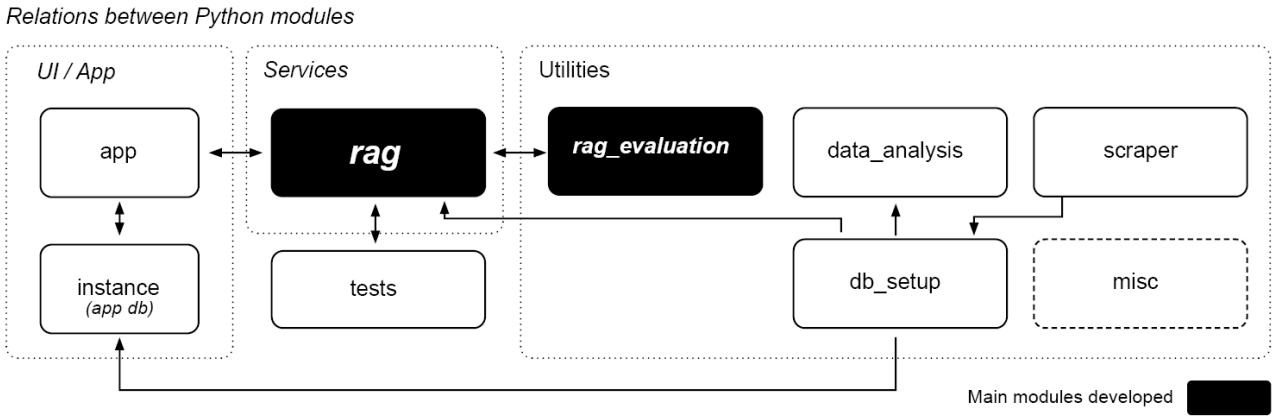


Fig 3.2 - System design: Prototype high-level diagram

Group	Module name	Responsibility
UI / app	*app*	Handles the UI and SQL database.
	instance	The SQL database, coordinated with the RAG vector database.
Services	*rag*	The RAG service and its vector database, able to take queries and return results.
	tests	A dedicated, simple testing suite for the RAG service only.
Utilities	*rag_evaluation*	The evaluation suite for assessing RAG performance.

scraper	A helper module to scrape secondary data.
data_analysis	A module to perform an initial exploratory data analysis on scraped data.
db_setup	A helper module to setup the SQL database using scraped data.

Table 3.2 - Module summary

3.3 Data collection and processing (data pipeline implementation)

Two main types of data have been used on the project, in line with the ethics form:

A. Data for RAG core database: qualitative (Hassan, 2025), secondary data - main data corpus consisting of about forty (40) blog articles, considered unseen by the local LLM due to either: being released to the public domain after its cutoff date, or the prompt used constrains the answers to a provided context.

B. Data from test users: Primary data from a short survey filled out by test users on **Microsoft Forms** (Microsoft, 2025) for qualitative evaluation.

Figure 3.3 shows the data pipeline conceptual diagram:

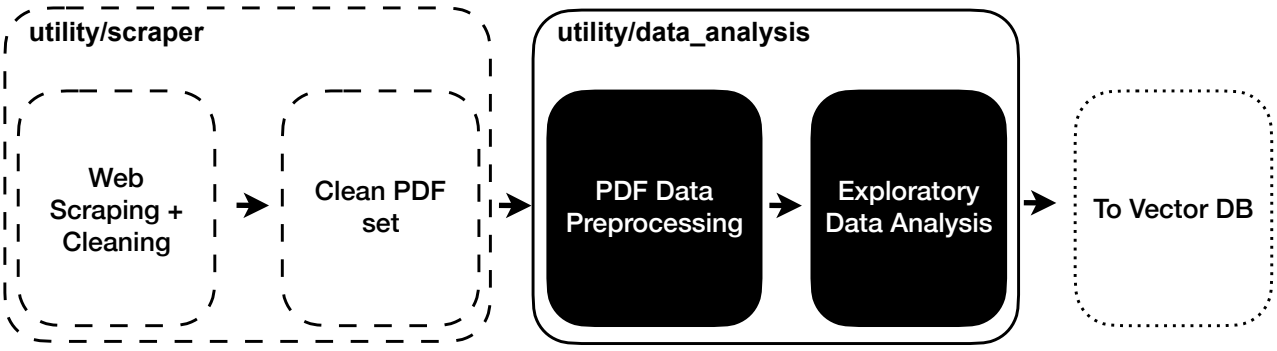


Fig 3.3 - Data pipeline methodology diagram

3.3.1. Data sources

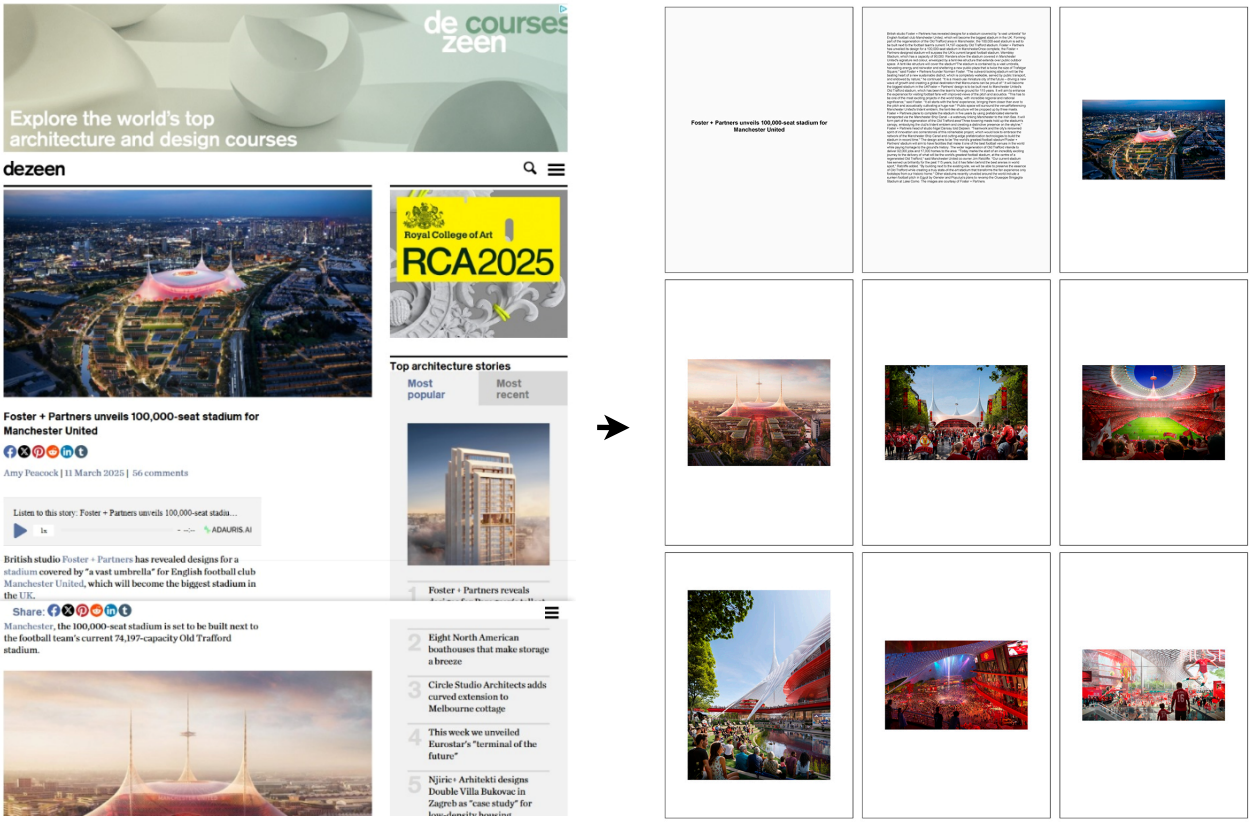
The main secondary data source used is an architectural design specialised blog, Dezeen.com (Dezeen, 2025) One example of a typical article is: “https://www.dezeen.com/2025/03/11/manchester-united-stadium-foster-partners” (Peacock, 2025).

3.3.2 Data collection

Secondary data analogous to real-world documents has been programatically collected from a set of URLs via web scraping using the Python package **BeautifulSoup** (Richardson, 2025), returning clean text and images from the main content.

3.3.3 Data cleaning and preprocessing

Raw scraped data has been formatted into A4 PDF documents using **Reportlab** (ReportLab, 2025). The PDF files are the corpus that will be used by the RAG and exploratory data analysis.



Screenshot of original data source (web page)

Clean PDF file containing core text and images only (full scale version on Appendix)

Fig 3.3.3 - Original web scraped vs converted PDF

3.3.4. Additional public dataset used for benchmarking

An additional secondary data dataset, RAGBench (Friel, Bely and Sanyal, 2025), publicly available on **HuggingFace** (huggingface.co, n.d.) has also been used for benchmarking. A

small slice of 30 samples from its TechQA subset has been used to minimise computational load.

3.4 Exploratory data analysis

An exploratory data analysis has been done to have a general idea of the information and keyword distribution on the data. The results of this analysis are discussed on 4. Results.

3.4.1. Word cloud analysis

Using the **WordCloud** package (Mueller, 2025) has provided a visual representation of the most frequent words (excluding stop-words) in the text corpus.

3.4.2. Word count bar chart

In addition to 3.3.1, a bar chart displaying the top 50 words by frequency across the corpus has been done using the **Matplotlib** package (Matplotlib.org, 2025).

3.4.3. File/Word heat map

A heat map chart showing occurrence of each of the top 50 words (X axis) against each file (Y axis) has been plotted to reveal potential hidden patterns.

3.4.4. Clustering words by topic / similarity

To help reveal underlying structures from the previous heat map on 3.3.3, further processing has been done consisting of embedding the original list of words with **Sentence Transformers** (huggingface.co, 2025), and clustering them using **KMeans** (scikit-learn.org, 2025).

3.5 RAG module design

Based on the literature review, a set of baseline architectures has been initially proposed, then evolved into an advanced architecture that tries to take advantage of the main enhancement opportunities identified. The shortlisted architectures are discussed below:

3.5.1. Baseline architecture

A baseline architecture consisting of essential elements, curated based on literature review, such as data pre-processor, vector database, retriever and generator components

has been laid out considering suitable technologies like **ChromaDB** (Chroma, 2025), **LangChain** (LangChain, 2025) and **Ollama** (Ollama, 2025). For the case of image and multimodal RAG types, additional layers for image manipulation, VQA and caption generation have been added, using frameworks such as **Pillow** (Pillow, 2025), **Pytorch** (Pytorch, 2025) and **Transformers** (huggingface.co, 2025).

3.5.2 Advanced architecture

A number of opportunities for enhancement identified on the literature have been explored and implemented on the solution, on top or as a complement to the baseline architecture. These revolve around some of the core items, as shown on figure 3.5.2.

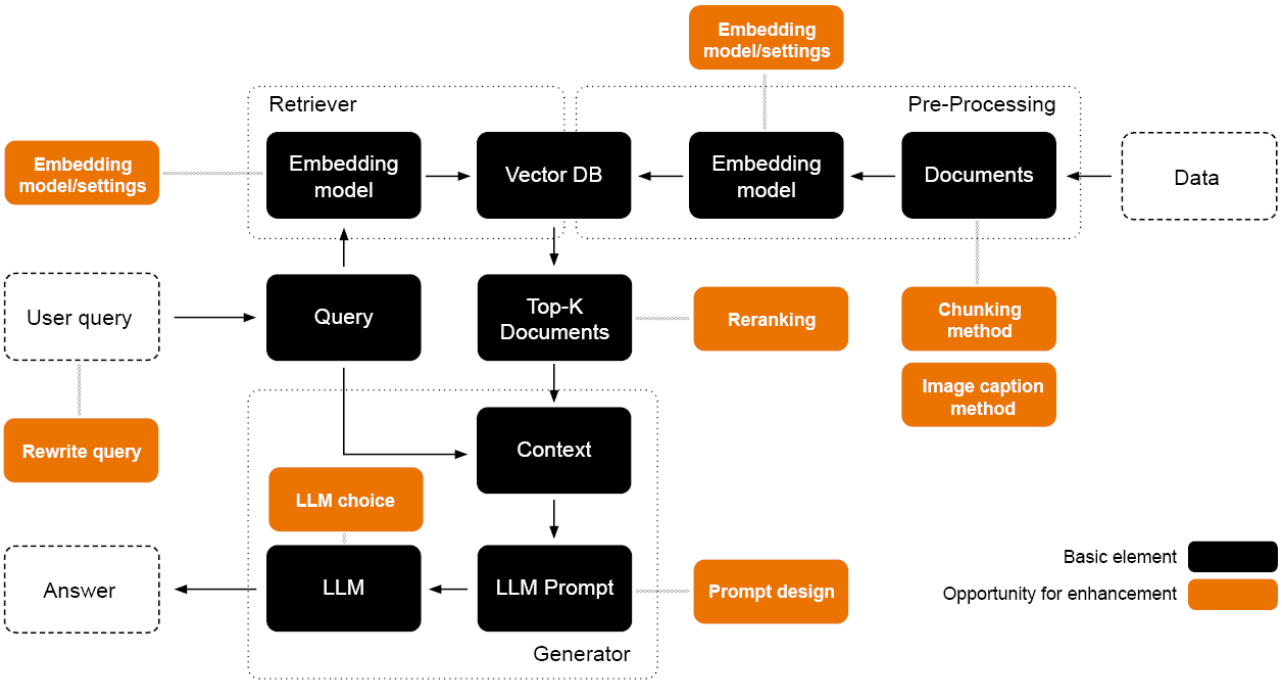


Fig 3.5.2 - Actual RAG enhancements tested - Adapted from (huggingface, co, n.d.) (IBM Retrieval Augmented Generation, n.d.)

In addition to the frameworks already used for baseline setups, **Sentence Transformers** (huggingface.co, 2025) has been used for the re-ranking feature.

3.6 RAG module implementation

The core module of this research, comprises a set of classes that rely on the AI and NLP frameworks mentioned on 3.5.1/2. It is designed as a standalone module, acting as a

service to a hypothetical broader "app" (limited to just the UI for this scope). A summary of the implementation is highlighted below (similar methodology applies to all other modules):

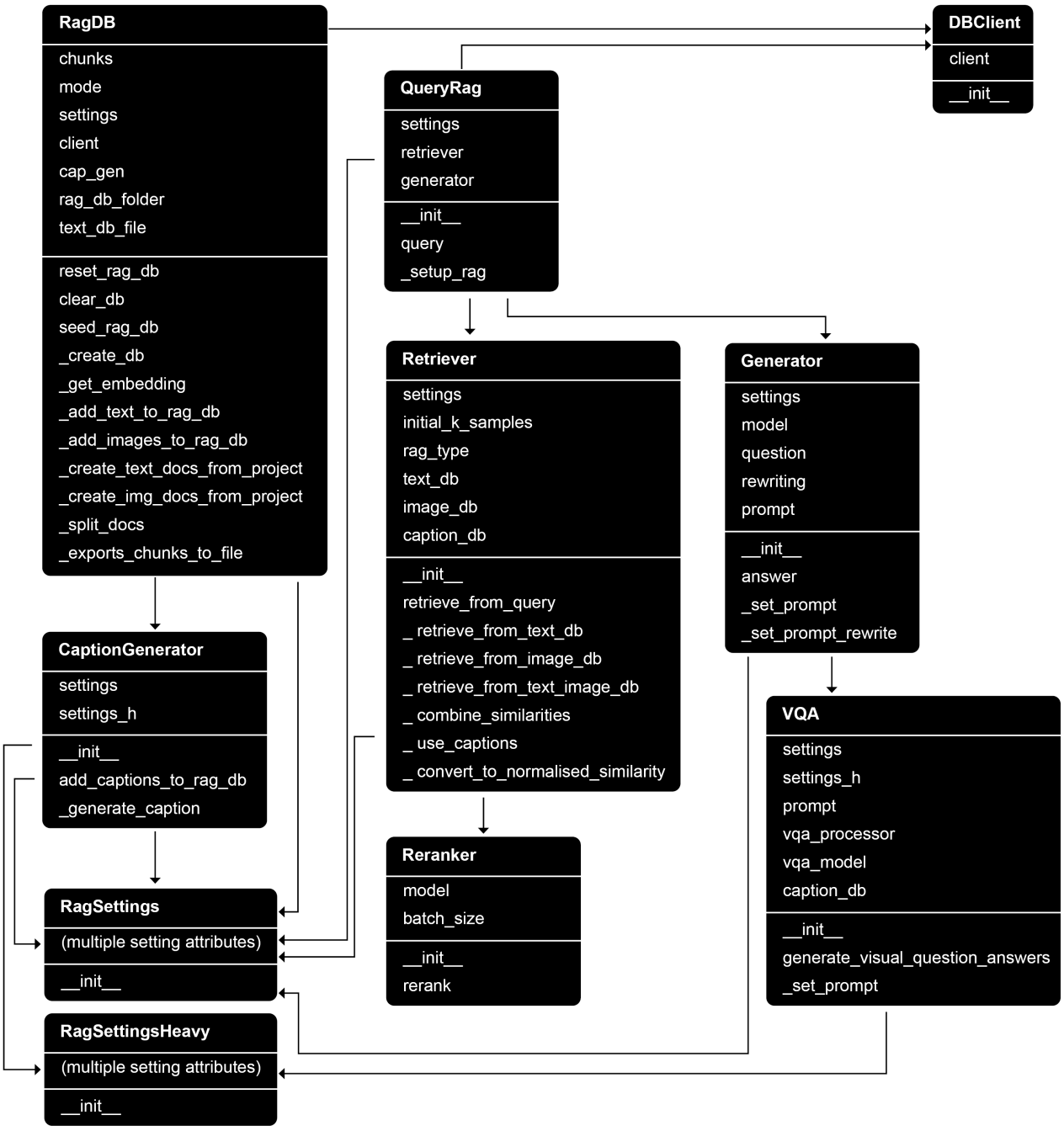


Fig 3.6a - A (simplified) UML diagram of the RAG module

Class name	Description
DBClient	Wrapper to Chroma DB client to access vector DB with mode configuration.
QueryRag	Handler to perform RAG queries.
Retriever	Accesses the DB and returns relevant results from queries.
Generator	Generates answers from input queries. Also acts as a rewriter.
Reranker	Takes initial results and re-ranks by resorting them by similarity.
VQA	Queries images content from input query on demand.
RagDB	Builds and updates the RAG vector database.
CaptionGenerator	Generates captions and populates caption_db collection on the vector DB.
RagSettings	Stores the various parameters of the RAG setup.
RagSettingsHeavy	Stores heavy parameters used on image and multimodal RAG only.

Table 3.6b - RAG module class summary

3.6.1 Test suite

An essential unit testing suite has been developed, strictly limited to the RAG module classes, to check:

- System stability after refactoring or upgrades during development.
- Validation of output and value types returned.

3.7 User interface implementation (the "App")

A user interface has been developed using the Python framework **Flask** (Pallets Projects, 2025) with **Tailwind CSS** (Tailwind CSS, 2025) and *Prettier* (Prettier, 2025) applied to Tailwind only for maintainability purposes, **SQLAlchemy** (SQLAlchemy, 2025) for UI database handling and a **JavaScript** (ECMA International, 2025) helper module that refreshes the page upon new requests.

Figures 3.7a and 3.7b show two screenshots of the user interface before and after answering a query:

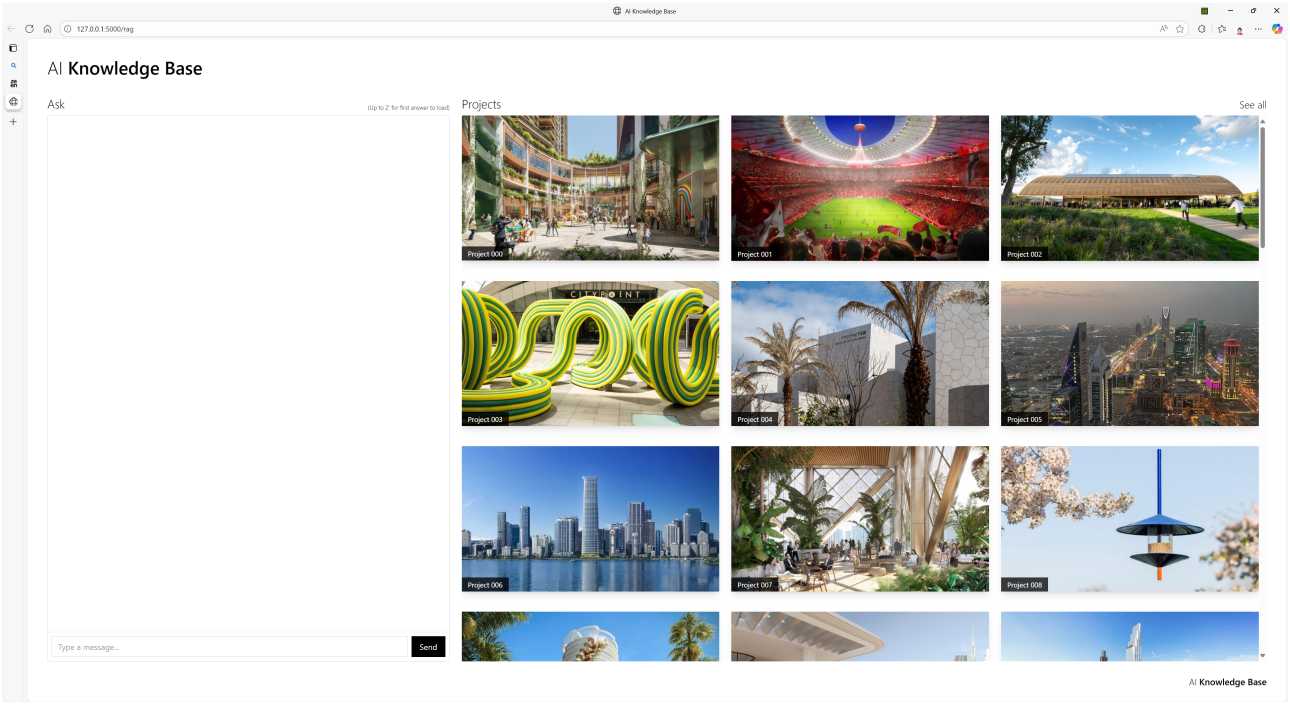


Fig 3.7a- Prototype UI showing all projects and the chat box ready to answer queries

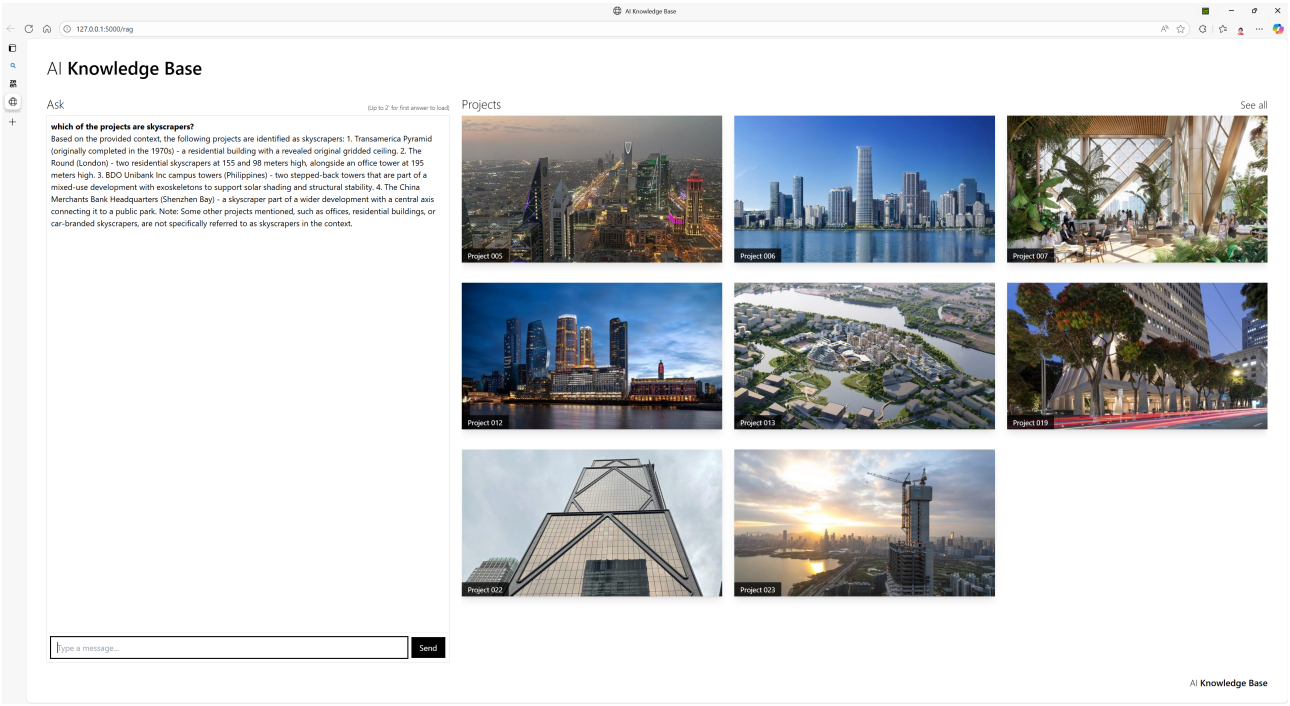


Fig 3.7b Screenshot of the UI, after answering a query (project grid refreshes according to response)

3.8 Evaluation suite module implementation

A quantitative evaluation module acts as a key piece of the research. By running multiple configurations on the RAG module, it helps identify the main elements that impact RAG performance.

The three main RAG types evaluated are: Text-Only, Image-Only, and Multimodal (Text + Image), from which a baseline model and a best performer have been compared:

Capability	Development level	Code name
Text-Only	Baseline	A1
	Fine-tuned (best performer)	A2
Image-Only	Baseline	B1
	Fine-tuned (best performer)	B2
Multimodal (Text + Image)	Baseline	C1
	Fine-tuned (best performer)	C2

Table 3.8 - Types of RAG tested

A list of questions with their respective ground truth answers, organised according to key categories, has been used. A set of key metrics has been identified and tracked through the experiments.

A systematic fine-tuning approach has been taken. After initially using **ParameterGrid** (scikitlearn.org, 2025), this method was eventually been replaced with a specialist optimisation package, **Optuna** (Optuna.org, 2025), with greater functionality. The evaluation has been run with multiple sets of parameters using the **RagSettings/ RagSettingsHeavy** classes in **three steps**, as shown on chapters 4.2, 4.3 and 4.4 and introduced below:

3.8.1. Step 0 (Baseline run)

All parameters set to baseline settings (baseline values are attached on Appendix) for text, image and text + image modes.

3.8.2. Step 1 (First pass)

Permutations on just a few major/key parameters (in **bold**), keeping the rest of the search space unchanged to baseline value. This prevents unnecessary computation time at this stage, as follows:

Parameter	**Values** (full disclosure of values on appendix)	**Description**
rag_type	**[TEXT, IMAGE, TEXT_IMAGE]**	**The RAG mode.**
model_choice	[BASELINE_LLM]	Particular LLM to use.
text_embeddings_model	[BASELINE_TEXT_EMBEDDING_MODEL]	Text embeddings model.
initial_k_samples	[BASELINE_INITIAL_K_SAMPLES]	Pre-reranking K samples.
k_samples	[BASELINE_K_SAMPLES]	Final K samples
text_weight	[BASELINE_TEXT_WEIGHT]	Text weighting for multimodal.
img_weight	[BASELINE_IMG_WEIGHT]	Image weighting for multimodal.
caps_weight	[BASELINE_CAPS_WEIGHT]	Caption weighting for multimodal.
batch_size	[BASELINE_BATCH_SIZE]	Generation batch size.
min_new_tokens	[BASELINE_MIN_NEW_TOKENS]	Minimum number of tokens generated by LLM.
max_new_tokens	[BASELINE_MAX_NEW_TOKENS]	Maximum number of tokens generated by LLM.
decoding	[BASELINE_DECODING]	Decoding strategy.
num_beams	[BASELINE_NUM_BEAMS]	Number of beams in beam search decoding.
no_repeat_ngram_size	[BASELINE_NO_REPEAT_NGRAM_SIZE]	Repetition of N grams in output.
length_penalty	[BASELINE_LENGTH_PENALTY]	Penalty to longer or shorter outputs.
num_io_workers	[BASELINE_NUM_IO_WORKERS]	Concurrent I/O workers on loading data.
cache_answers	[BASELINE_CACHE_ANSWERS]	Caching of previous answers.
pre_compute_captions	**[BASELINE_CAPTION_FLAG, False]**	**Consider image captions for context (baseline=true).**
use_reranker	**[BASELINE_USE_RERANKER, True]**	**Use reranker functionality (baseline=false).**
reranker_model	[BASELINE_RERANKER_MODEL]	Model used for reranking.
reranker_batch_size	[BASELINE_RERANKER_BATCH_SIZE]	Batch size for reranker model.
use_rewriter	**[BASELINE_USE_REWRITER, True]**	**Enable rewriting (baseline=false)**

rewriter_model	[BASELINE_REWRITER_MODEL]	Model for rewriting queries.
domain_hints	[BASELINE_DOMAIN_HINTS]	Keywords as context for generation.

Table 3.8.2 - RAG settings used on first pass (main features)

3.8.3. Step 2 (Second pass)

Some settings have been fixed according to results from the first pass (see 4. Results). Table 3.6.3 shows parameters selected for the second run based on estimated impact on performance (number of permutations is minimised to save computational resources):

Parameter	**Values** (full disclosure of values on Appendix)	**Description**
rag_type	*[TEXT]*	*Set to TEXT after first pass.*
model_choice	**[BASELINE_LLM, "mistral:latest"]**	**Particular LLM to use.**
text_embeddings_model	[BASELINE_TEXT_EMBEDDING_MODEL]	Text embeddings model.
initial_k_samples	*[BASELINE_INITIAL_K_SAMPLES]*	*Not using reranking.*
k_samples	**[BASELINE_K_SAMPLES, 5]**	**Final K samples**
text_weight	*[BASELINE_TEXT_WEIGHT]*	*Only text, no weighting needed.*
img_weight	*[BASELINE_IMG_WEIGHT]*	*Only text, no weighting needed.*
caps_weight	*[BASELINE_CAPS_WEIGHT]*	*Only text, no weighting needed.*
batch_size	**[BASELINE_BATCH_SIZE, 2]**	**Generation batch size.**
min_new_tokens	**[BASELINE_MIN_NEW_TOKENS, 50]**	**Minimum number of tokens generated by LLM.**
max_new_tokens	**[BASELINE_MAX_NEW_TOKENS, 300]**	**Maximum number of tokens generated by LLM.**
decoding	[BASELINE_DECODING]	Decoding strategy.
num_beams	**[BASELINE_NUM_BEAMS, 4]**	**Number of beams in beam search decoding.**
no_repeat_ngram_size	**[BASELINE_NO_REPEAT_NGRAM_SIZE, 3]**	**Repetition of N grams in output.**
length_penalty	**[BASELINE_LENGTH_PENALTY, 1.1]**	**Penalty to longer or shorter outputs.**
num_io_workers	[BASELINE_NUM_IO_WORKERS]	Concurrent I/O workers on loading data.
cache_answers	[BASELINE_CACHE_ANSWERS]	Caching of previous answers.
pre_compute_captions	*[False]*	*Set to false after first pass.*
use_reranker	*[False]*	*Set to false after first pass.*

reranker_model	*[BASELINE_RERANKER_MODEL]*	*Reranking not enabled.*
reranker_batch_size	*[BASELINE_RERANKER_BATCH_SIZE]*	*Reranking not enabled.*
use_rewriter	*[False]*	*Set to false after first pass.*
rewriter_model	*[BASELINE_REWRITER_MODEL]*	*Rewriter not enabled.*
domain_hints	**[BASELINE_DOMAIN_HINTS, ""]**	**Keywords as context for generation.**

Table 3.8.3 - RAG settings used on second pass (fine tuning of hyper parameters)

3.8.4. Evaluation questions

A short set of questions has been written and fed to the evaluation suite. These questions focus, mainly, on information held on the text corpus, thus there is a known bias towards text-based setups.

Category	Question sample	Expected answer sample
1. Numeric (quantitative information retrieval)	"How many meters tall is the 270 Park Avenue supertall skyscraper by Foster + Partners? (Answer with the number only)"	"423"
2. Architectural feature related (qualitative information retrieval)	"What pavilion has wooden seating? (Answer with less than 5 words)"	"London Festival of Architecture"
3. Location-related (qualitative information)	"In which city is the 18 Blackfriars Road development located? (Answer with the name of the city only)",	"London"

Table 3.8.4 - Some sample questions passed on to the RAG for evaluation

3.8.5. Quantitative metrics

Table 3.8.5a shows the set of quantitative metrics used. Precision@K and Recall@K have been eventually simplified to Hit@K to avoid the need for a significantly larger dataset, needing multiple golden passages to be supplied per question.

Type	Stage	Metric	Description
Quantitative metrics (Deepchecks Community Blog, 2024) (TruEra, 2024) (Kuang et al, 2025)	**Retrieval**	Hit@K	*Is the relevant item in top k?*
		Mean reciprocal rank (MRR)	*How fast the first relevant item appears in the list*
	Generation	BLEU (Bilingual Evaluation Understudy)	*Evaluates quality of generated text VS reference text*
		ROUGE (Recall-Oriented Understudy for Gisting Evaluation)	*Compares how much of the reference text appears on the generated text*
		METEOR (Metric for Evaluation of Translation with Explicit ORdering)	*Improves insights from BLEU by measuring semantic similarity.*
	Holistic	Groundedness	*Measures hallucinations on LLM output compared to retrieved text.*
		Answer relevance	*Confirms whether it is addressing the question correctly.*
		Context relevance	*Ensure each piece of retrieval is relevant to query.*
	Time	TTC (Time to complete)	*Measure time to complete a single query, from user input to LLM answer.*

Table 3.8.5a - Quantitative metrics used for evaluation

A weighting based on literature and adapted to this use case has been assigned to each metric, to form an aggregate **score**. The calculation prioritises Hit@K (it is critical that the golden reference is present among the results returned) and the holistic metric Groundedness (a broad quality metric), plus it sets a penalty on latency. Different weighting assignations would, of course, return different best performers.

Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Relevance	Context_Relevance	Avg_Time(s)
0.25	0.05	0.05	0.05	0.05	0.25	0.1	0.1	-0.1

Table 3.8.5b - Weighting of each of the metrics used relative to total score

3.9 Test user survey design

A small group of test users relevant to the field of architectural design has assessed their impressions on the prototype. They have been able to use it for a short time, and have

answered a set of questions posted on **Microsoft Forms** (Microsoft, 2025), based on the criteria discussed on table 3.9.1 and shown on table 3.9.2.

3.9.1. Qualitative metric criteria

Table 3.9.1 shows the criteria for assessment of qualitative metrics. A survey for test users of the quantitative best performers has been conducted on the basis of these criteria on 3.9.2:

Type	**Stage**	**Metric**	***Description***
Qualitative metrics (Chen et al., 2025)	**User survey**	Subjective impressions on prototype - questions adapted from standard System Usability Scale (Brooke, 1995) questions	*Survey with overall user experience ("usefulness" perception) of the system, compared to legacy search or peer to peer exchange. Each question has its individual rating/score and an average general score has also been calculated.*

Table 3.9.1 - Qualitative metrics criteria

3.9.2. Survey questions

Table 3.9.2 shows a set of questions, which have been adapted from the standard *System Usability Scale* (Brooke, 1995) and tailored to make them more relevant to the use case presented in this research. Questions relative to time to completion and task completion have been omitted, since the former was explored in depth at the quantitative evaluation stage and the latter is already implicit in some of the first questions.

#	**Question**
1	How relevant were the text answers returned?
2	Did the text answers feel factually accurate?
3	Were the project images returned relevant to your question?
4	Did the combination of text and project images help?
5	When you asked some question where the information was not present on the text description, how accurate was the response?
6	Was the answering speed acceptable?
7	Do you think a tool like this could be useful as a way to find key information about projects?

8	How useful do you think this tool could be (provided projects have a reasonably good text description), compared to, i.e. just manually finding files with Windows Explorer?
9	What additional feature(s) or changes do you think would make this tool more useful than it is right now?
10	Do you imagine any other potentially useful applications of this tool, or an evolved version of it, other than knowledge management? (i.e. marketing, visual communication, etc.)

Table 3.9.2 - Test user survey questions

3.10 Limitations of this methodology

There are some limitations and known biases inherent to the methodology used:

- The set of questions and ground truth passed to the RAG are based on the text corpus only - this will penalise image-only results and it is a known bias of this study, although experimental testing with the means available has proven such captions to hardly be as descriptive and helpful as the text.
- Real-time visual question answering performance might greatly benefit from upgraded hardware, not available for this study.
- Since focus has been on usability and performance aspects, emphasis on software robustness, like error handling, has been limited.

3.11 Reflections

All methodology has been based on a practical approach within an ethical perspective, with all data used to feed the RAG being secondary, collected from the public domain. Test user survey data is anonymous and has been appropriately processed, according to standard university procedure.

4 RESULTS AND ANALYSIS

Starting from the the data analysis, and followed by the RAG metrics from the various hyper parameter configurations tested, all results from the experiments discussed on the previous chapter are analysed on the following sections.

4.1 Exploratory Data Analysis (EDA) results

Results of the exploratory data analysis introduced on 3.4 that have been used to better understand the content on the data are displayed below.

4.1.1. Word cloud analysis

Figure 4.1.1 features the resulting word cloud from all combined (text) content. As expected, it becomes clear that the corpus holds architectural design, given the returned cloud:

Fig 4.1.1 - Word cloud analysis

4.1.2. Word count bar chart

Likewise, a bar chart displaying the top 50 words by frequency across all PDF files providing a clearer overview of the actual word count for each term:

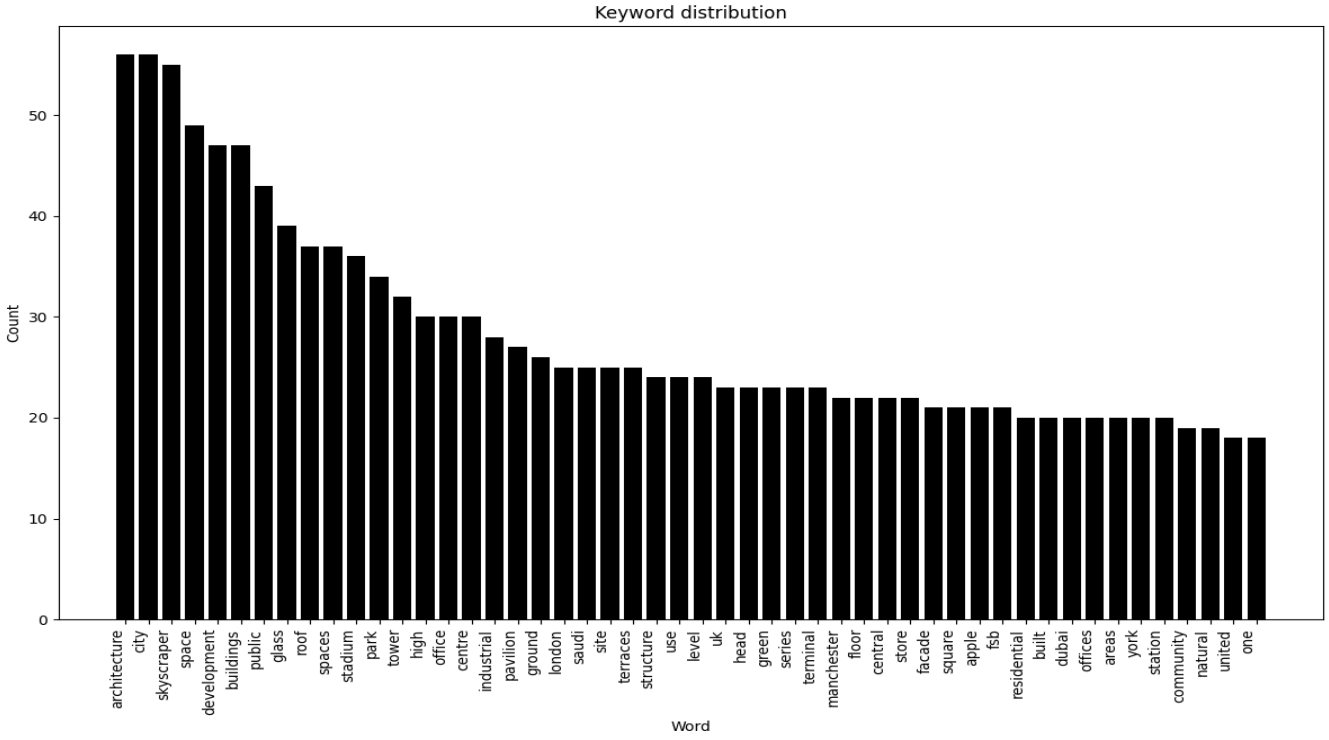


Fig 4.1.2 - Word count bar chart

4.1.3. File / word heat map

File/Word heat map results, as discussed on 3.4.3 show a noisy heat map with no obvious underlying pattern on fig. 4.1.3:

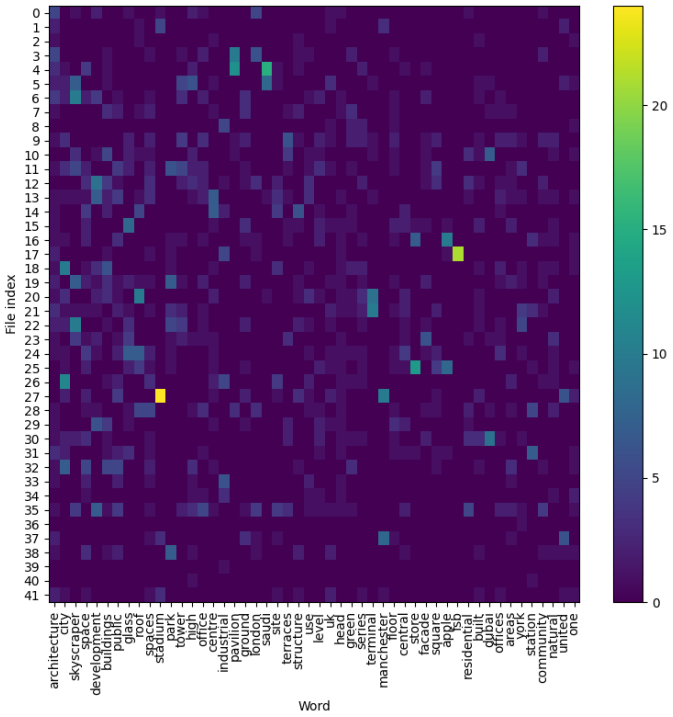


Fig 4.1.3 - File / word heat map before clustering

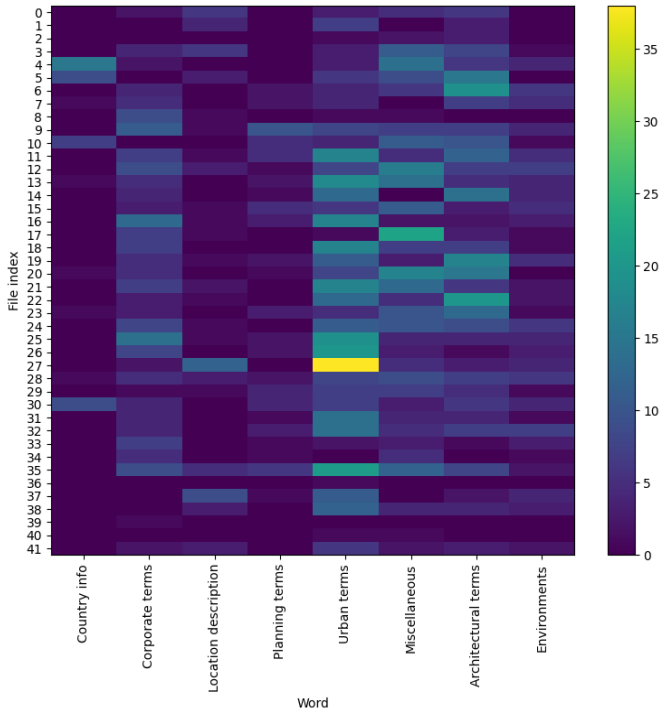


Fig 4.1.4 - Word clusters by topic / similarity

4.1.4. Clustering words by topic / similarity

After applying a clustering algorithm following methodology discussed on 3.4.4, a more clear underlying structure emerges on fig. 4.1.4, surfacing broader topics related to urban design and architectural design jargon.

4.1.5. Insights from the exploratory data analysis

Although further thorough data cleanup could be performed, the analysis conducted already gives a good overview of the main topics present on the data. Some insights are highlighted below:

- The original heat map showed a disperse distribution with no clear trends nor strong correlations
- After word embeddings and clustering are applied, more meaningful patterns emerge, as shown on the second heat map.
- The second heat map shows what broad topics are mentioned in which documents, and their indexes, which could be a valuable high-level insight when trying to understand patterns in large, real-world document datasets.

This analysis has been used to **underpin the topics of the questions fed to the RAG** for quantitative evaluation from 3.8.4.

4.2 Step 0 of RAG hyper parameter fine tuning: baseline run

Baseline RAG system performance evaluation results shown on table 4.2 below. Best performers are text-only modes. Image-only mode lags behind on almost all metrics, with much higher inference times, suggesting how image-based modes might be more complex to effectively implement than text-only.

Mode	Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Relevance	Context_Relevance	Avg_Time(s)	Score
A1 (text)	1	**0.22**	**0.08**	**0.29**	0.16	0.98	0.50	**0.85**	**0.98**	**0.76**
B1 (image)	0	0.01	0.11	0.19	**0.17**	0.35	**0.68**	0.20	8.07	0.11

C1 (text_image)	1	**0.22**	0.01	**0.29**	0.15	**1.00**	0.54	0.83	8.75	0.62

Table 4.2 - Baseline RAG evaluation results

4.3 Step 1 of RAG hyper parameter fine tuning: first pass

Results from the first pass permutations can be seen on table 4.3.1. The table has been sorted by score, indicating the iteration number on the left index column.

#	Mode	Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Relevance	Context_Relevance	Avg_Time (s)	Score
23	text	**1**	**0.22**	**0.10**	**0.32**	0.17	0.95	0.57	0.83	**0.52**	**0.82**
3	text	**1**	**0.22**	0.00	0.31	0.15	**1.00**	0.53	**0.87**	3.01	0.75
20	text	**1**	**0.22**	0.00	0.25	0.14	**1.00**	0.52	0.85	0.54	0.75
19	text	**1**	**0.22**	0.08	0.22	0.27	0.72	**0.67**	**0.87**	2.91	0.74
13	text	**1**	**0.22**	0.01	0.28	0.14	**1.00**	0.49	**0.87**	2.39	0.74
4	text_image	**1**	**0.22**	0.01	0.29	0.14	0.97	0.56	**0.87**	8.26	0.72
9	text_image	**1**	**0.22**	0.05	0.27	0.14	0.97	0.47	**0.87**	8.45	0.70
5	text	**1**	**0.22**	0.06	0.16	0.26	0.72	0.57	**0.87**	1.89	0.70
2	text_image	**1**	**0.22**	0.01	0.29	0.14	1.00	0.53	**0.87**	13.84	0.68
6	text	**1**	**0.22**	0.09	0.20	0.28	0.70	0.45	0.80	1.56	0.68
1	text_image	**1**	**0.22**	0.01	0.28	0.14	**1.00**	0.52	**0.87**	13.48	0.68
16	text_image	**1**	**0.22**	0.07	0.21	**0.29**	0.77	0.52	0.82	7.97	0.67
0	text_image	**1**	**0.22**	0.08	0.23	0.30	0.67	0.42	**0.87**	15.75	0.58
7	text_image	**1**	**0.22**	0.08	0.21	0.24	0.72	0.41	**0.87**	15.65	0.57
8	image	0	0.01	**0.10**	0.19	0.18	0.53	0.61	0.23	8.07	0.22
11	image	0	0.01	0.03	0.11	0.13	0.33	0.52	0.25	7.58	0.07
14	image	0	0.01	0.03	0.11	0.14	0.43	0.40	0.20	7.58	0.06
15	image	0	0.01	0.08	0.09	0.09	0.45	0.46	0.13	13.01	0.05

10	image	0	0.01	**0.10**	0.15	0.17	0.28	0.49	0.13	13.02	0.04
17	image	0	0.01	0.03	0.11	0.16	0.38	0.41	0.13	13.68	0.00
18	image	0	0.01	0.02	0.10	0.14	0.42	0.35	0.13	14.09	-0.02
12	image	0	0.01	0.10	0.17	0.16	-	0.54	0.20	7.13	-
21	text_image	**1**	**0.22**	0.08	0.18	0.28	-	0.65	0.85	7.97	-
22	text	**1**	**0.22**	0.07	0.18	0.24	-	0.55	**0.87**	3.05	-

Table 4.3.1 - First pass results

Extracting the best performers on each category, table 4.3.2 shows a comparison between the baseline models and the best performers on each modality, showing higher scores for all fine-tuned models after the first pass:

Mode	Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Relevance	Context_Relevance	Avg_Time(s)	Score
A1 (text)	**1**	**0.22**	0.08	0.29	0.16	**0.98**	0.50	0.85	0.98	0.76
A2 (text)	**1**	**0.22**	**0.10**	**0.32**	**0.17**	0.95	0.57	0.83	**0.52**	**0.82**
B1 (image)	0	0.01	**0.11**	0.19	**0.17**	0.35	**0.68**	0.20	8.07	0.11
B2 (image)	0	0.01	0.10	0.19	0.18	0.53	0.61	0.23	8.07	0.22
C1 (text_image)	**1**	**0.22**	0.01	**0.29**	0.15	**1.00**	0.54	0.83	8.75	0.62
C2 (text_image)	**1**	**0.22**	0.01	0.29	0.14	0.97	0.56	**0.87**	8.26	0.72

Table 4.3.2 - Comparison between baseline and best performers after first-pass iterations

From this first pass, and considering the limitations of this study, the highest score corresponds to the text-only mode RAG #23, hence it has been selected as a base setup for further fine-tuning of second-order hyper parameters.

Fig. 4.3.1 shows a three-dimensional plot of the results from table 4.3.3:

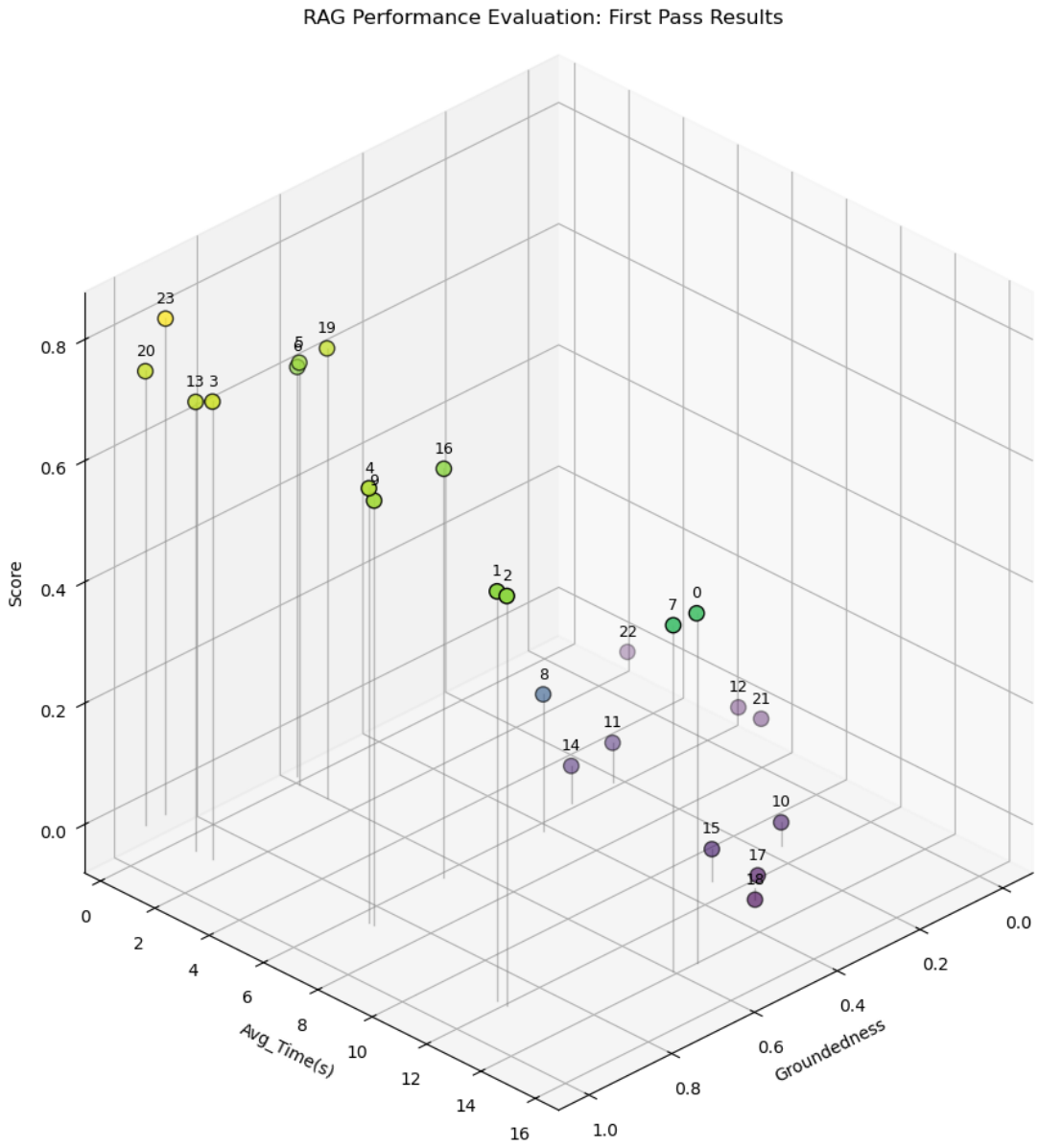


Fig 4.3.1 - Results plot from first pass

From the settings used as search space on the first pass evaluation run, #23 had the setup shown on table 4.3.4:

Parameter	**Values** (full disclosure of values on appendix)	**Description**
rag_type	[TEXT]	RAG mode set to text-only.
pre_compute_captions	[False]	Ignore captions database.
use_reranker	[False]	Reranker disabled.
use_rewriter	[False]	Rewriting disabled.

Table 4.3.4 - First pass best performer's setup

By looking at table 4.3.4, it appears that the additional enhancements (use of captions, reranking, rewriting) might have had some impact on individual metrics but not much on overall system performance. Regarding captions, this was expected, since the evaluation questions did not incorporate caption information - however, reranking and rewriting features had been expected to be more impactful that they actually were. The above settings have been set as fixed for a second, fine-tuning evaluation run.

4.4 Step 2 of RAG hyper parameter fine tuning: second pass

After this second pass, permutation #302 comes out on top. Table 4.4a shows the first ten rows of the sorted data frame. Note that scores across the different evaluation tables have been normalised and are run-dependent, therefore the scales vary across runs 0, 1 and 2, however actual performance has been be improving (since subsequent runs still include some of the setups from previous runs).

#	Mode	Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Relevance	Context_Relevance	Avg_Time(s)	score
302	text	**1**	**0.22**	0.03	0.34	0.17	**1.00**	**0.59**	**0.87**	0.54	**0.51**
395	text	**1**	**0.22**	0.03	0.34	0.17	1.00	0.52	0.83	0.44	0.44
309	text	**1**	**0.22**	0.08	0.33	0.17	0.98	0.56	0.83	0.43	0.43
397	text	**1**	**0.22**	0.06	0.32	0.16	1.00	0.47	**0.87**	0.51	0.43
315	text	**1**	**0.22**	0.08	0.34	0.17	0.98	0.53	0.83	0.51	0.43
49	text	**1**	**0.22**	**0.13**	**0.35**	**0.19**	1.00	0.52	0.80	0.88	0.42
433	text	**1**	**0.22**	0.01	0.32	0.15	1.00	0.56	0.83	**0.40**	0.42
338	text	**1**	**0.22**	0.12	0.33	0.18	1.00	0.56	0.80	0.88	0.41
414	text	**1**	**0.22**	0.05	0.33	0.17	1.00	0.52	0.83	0.79	0.41

162	text	**1**	**0.22**	0.04	0.33	0.17	1.00	0.50	0.83	0.46	0.41

Table 4.4a - Results from hyper parameter fine tuning from evaluation second pass

The top performer of this second pass has used the below setup - parameters with values other than baseline are **highlighted in bold**:

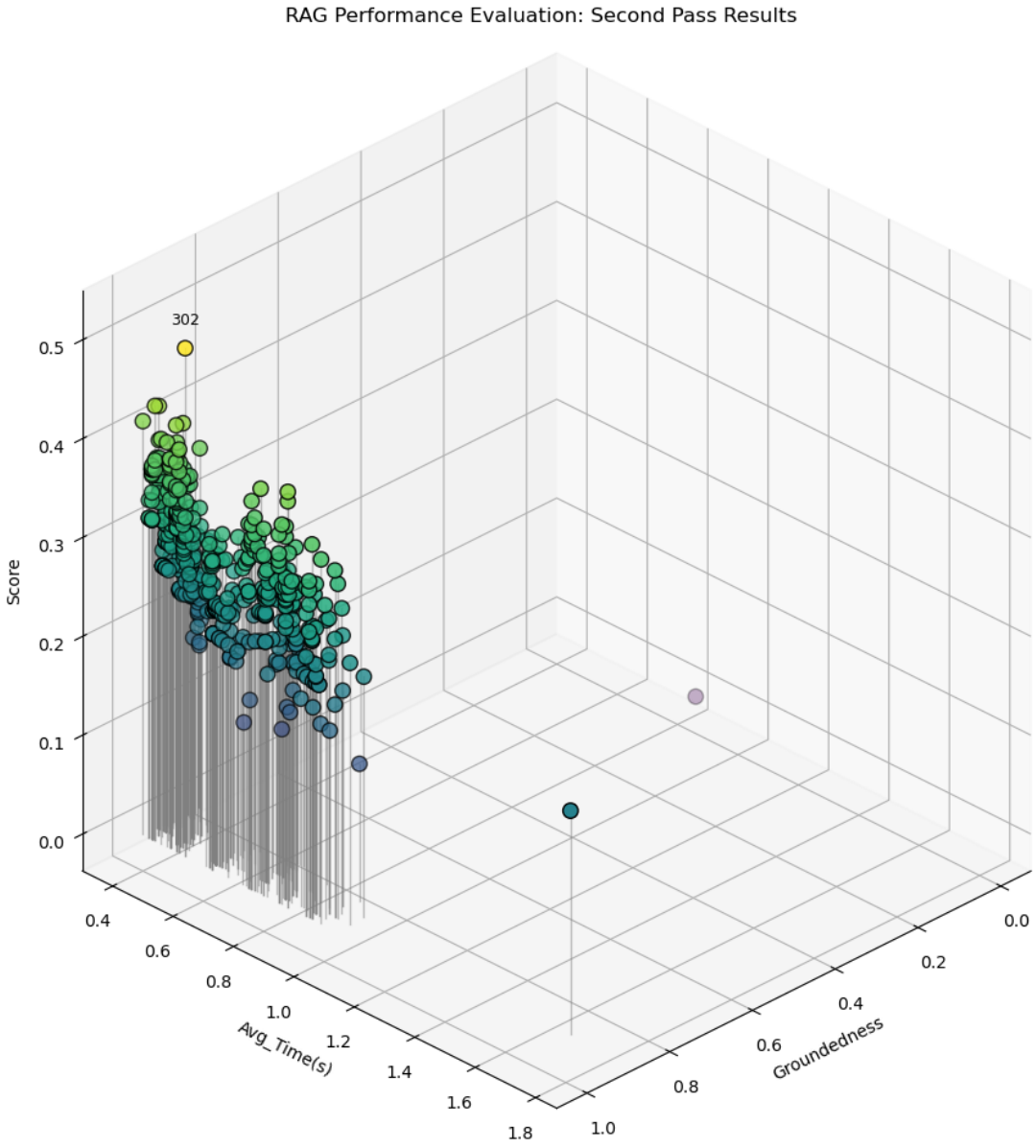


Figure 4.4 - Results plot from second pass

Parameter	**Values** (full disclosure of values on appendix)	**Description**
rag_type	[TEXT]	Set to TEXT after first pass.
model_choice	[BASELINE_LLM]	Particular LLM to use.
text_embeddings_model	[BASELINE_TEXT_EMBEDDING_MODEL]	Text embeddings model.
initial_k_samples	[BASELINE_INITIAL_K_SAMPLES]	Not using reranking.
k_samples	[BASELINE_K_SAMPLES, 5]	Final K samples
text_weight	[BASELINE_TEXT_WEIGHT]	Only text, no weighting needed.
img_weight	[BASELINE_IMG_WEIGHT]	Only text, no weighting needed.
caps_weight	[BASELINE_CAPS_WEIGHT]	Only text, no weighting needed.
batch_size	**[2]**	Generation batch size.
min_new_tokens	**[50]**	Minimum number of tokens generated by LLM.
max_new_tokens	[BASELINE_MAX_NEW_TOKENS]	Maximum number of tokens generated by LLM.
decoding	[BASELINE_DECODING]	Decoding strategy.
num_beams	**[4]**	Number of beams in beam search decoding.
no_repeat_ngram_size	[BASELINE_NO_REPEAT_NGRAM_SIZE]	Repetition of N grams in output.
length_penalty	**[1.1]**	Penalty to longer or shorter outputs.
num_io_workers	[BASELINE_NUM_IO_WORKERS]	Concurrent I/O workers on loading data.
cache_answers	[BASELINE_CACHE_ANSWERS]	Caching of previous answers.
pre_compute_captions	**[False]**	Set to false after first pass.
use_reranker	[BASELINE_USE_RERANKER]	Set to false after first pass.
reranker_model	[BASELINE_RERANKER_MODEL]	Reranking not enabled.
reranker_batch_size	[BASELINE_RERANKER_BATCH_SIZE]	Reranking not enabled.
use_rewriter	[BASELINE_USE_REWRITER]	Set to false after first pass.
rewriter_model	[BASELINE_REWRITER_MODEL]	Rewriter not enabled.
domain_hints	**[""]**	No domain hints passed.

Table 4.4b - Second pass best performers's setup

The set of best performing parameters after the second pass, from table 4.4b, has been selected for implementation ("production" settings) on the test user evaluation stage.

4.5 Benchmarking: testing the best performer against the RAGBench dataset (Friel, Belyi and Sanyal, 2025)

4.5.1 Benchmarking with first pass settings

As a trial, the text-only subset of the initial first pass permutations have been used against the small slice discussed on 3.3.4 from RAGBench. Table 4.5.1 shows the results:

#	rag_type	Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Rel.	Context_Rel.	Avg_Time (s)
0	text	**1**	**0.24**	**0.06**	**0.22**	**0.13**	-	0.51	0.12	5.06
1	text	**1**	0.24	0.01	0.17	0.10	**0.38**	**0.48**	**0.13**	4.52
2	text	**1**	0.24	0.02	0.17	0.09	-	0.49	0.05	**1.90**
3	text	**1**	0.24	0.04	0.21	0.12	0.33	0.57	0.02	1.98

Table 4.5.1 - Benchmarking results using first pass settings (text-only)

Context relevance value returned using RAGAS is consistent with the same from RAGBench (Friel, Belyi and Sanyal, 2025) ("RAGAS Rel" = **0.12**, table 3, page 9), indicating the methodology and solution proposed in this study might be aligned with those from other studies.

4.5.2 Benchmarking with second pass settings

This step has been skipped, do to the high number of permutations and computational load required.

4.5.3 Benchmarking with production settings

This run using production settings extracted from 4.4, table 4.5.3 shows a drop in some of the performance metrics, such as context relevance, against the benchmarking dataset. This suggests that **different datasets** (even of the same modality, i.e. text-only) **might need different settings** - making generalisation challenging for heterogeneous datasets.

rag_type	Hit@K	MRR	BLEU	ROUGE	METEOR	Groundedness	Answer_Rel.	Context_Rel.	Avg_Time (s)
text	1	0.24	0.05	0.20	0.12	0.32	0.53	0.018	2.59

Table 4.5.3 - Benchmarking results using production settings

4.5.4 Benchmarking summary

A summary of context relevance values obtained, compared to the same from the RAGBench (Friel, Belyi and Sanyal, 2025) on the RAGBench dataset:

rag_settings	Context_Rel.
first pass settings	**0.13**
production settings	0.018
RAGBench	0.12

Table 4.5.4 - Benchmarking results comparison against RAGBench

4.6 Survey results

The survey on first impressions on the prototype was shared with small but relevant number of professionals (twelve) in the field of architecture. Results of the survey on figure 4.6 show a reasonably well received prototype. Results website can be visited on **this link**. *Note: the two negative answers to the first question were due to form testing.*

Overall potential "usefulness" perception from the survey seems high (an average of **4.09/5** for the rated questions), and lowest scores where given to question 6 (lack of available context returns poorer responses). Some of the feedback received included:

What additional feature(s) or changes do you think would make this tool more useful than it is right now?

> *"Citations showing which part of the text description or image was used to generate the answer."*
> *"Depth and breadth of knowledge in the projects, currently feels like you need to know what information is available before asking."*

"Coordination between images and generated text."

Do you imagine any other potentially useful applications of this tool, or an evolved version of it? (i.e. marketing, visual communication, etc.)

"Often starting projects I want to find projects which may have a common feature e.g. towers in hot countries or ones that are naturally ventilated. This feels a useful way for sourcing information on precedents."
"Project discovery and learning."
"Yes, this would be very useful as a tool to organise unstructured files in Windows Explorer - performing contextual searches, etc."

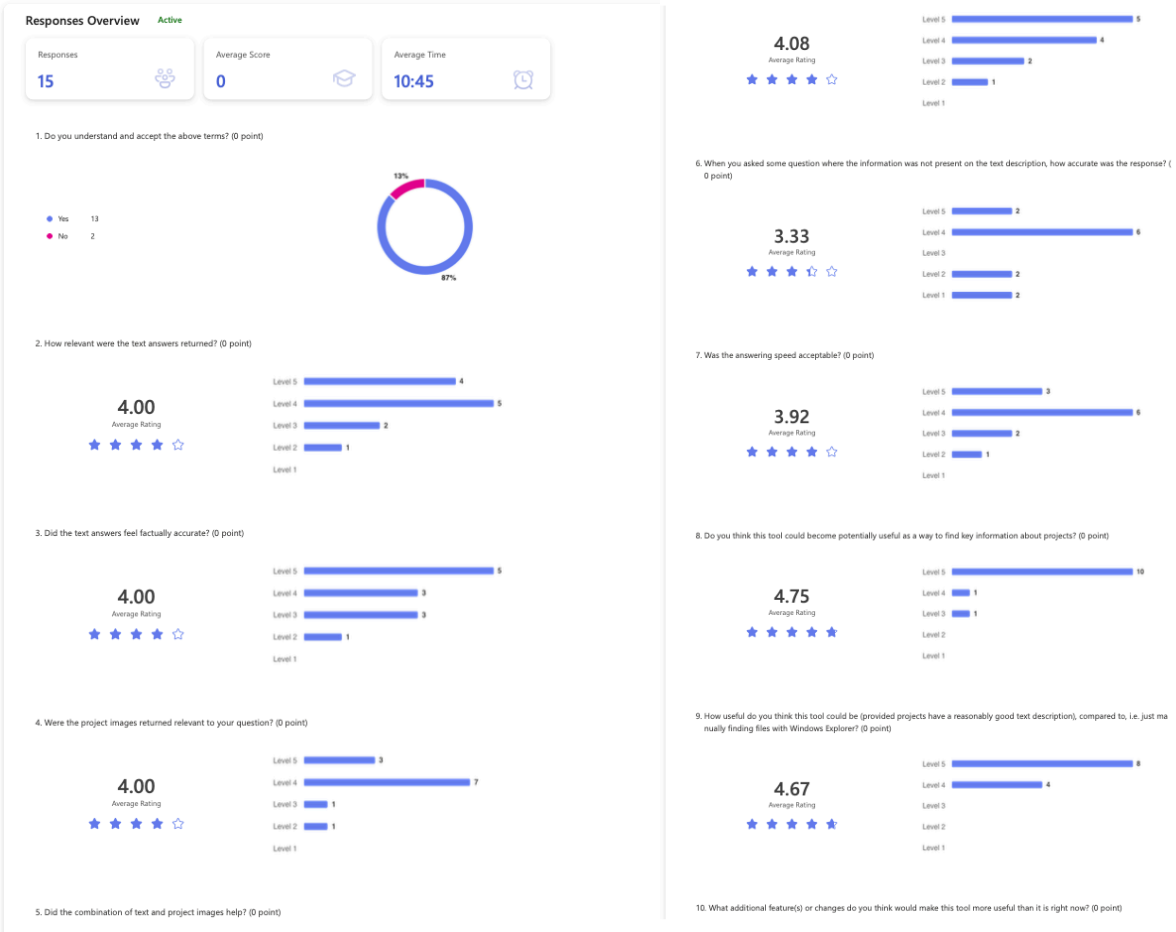


Fig. 4.6 - MS forms results page screenshots

4.7 Analysis and reflections on results

Both quantitative and qualitative analysis have thrown different, yet useful insights.

4.7.1 Quantitative results analysis

Considering hardware limitations, there seems to be a "sweet spot" where **a compromise can be found between latency and quality of the answers**. After a baseline configuration and a variety of more advanced configurations were tested, including re-ranking, re-writing, visual question answering and use of captions as context, the results have shown that **the use of advanced features might not be worth it for this use case**.

4.7.2 Quantitative results analysis

Despite the limited reach of the survey, it has shown some positive engagement. One of the most positive aspects has been the **contribution and ideas suggested by participants** for potential improvements of the tool, such as the coordination between text and images, as well as some of their thoughts confirming that such tool might become useful for use in KM for AD.

5 CONCLUSION AND RECOMMENDATIONS

This chapter discusses an overall evaluation of the research process conducted and its outcomes. Ethical, legal and professional considerations and personal reflections and recommendations for further research are also discussed below.

5.1 Critical evaluation of the research approach

While the approach taken has meant a **significant effort in product development**, this seemed the only way to really be able to effectively answer the research question. A back-end only prototype would not have been sufficient to gather qualitative insights from test users, while a quantitative-only approach did not seem to be enough to capture all the nuances and subjective feedback that could potentially be gathered from this research.

The prototype has been conveniently packaged, with a link provided with clear instructions **for reproducibility**, as discussed on the introduction to chapter 3.

5.2 Contributions of this research

This work aimed to answer the question: *Can a RAG system effectively retrieve and deliver expert-domain information, so that it could be a solution to knowledge management in large architectural practices?.* The outcome of this research has shown that **RAG could, indeed, be a viable tool for KM at large architectural practices**, provided:

- The **data ingested by the RAG has been previously curated to some extent**, so it contains key information within good text descriptions, which might be of help in the future (in particular, in text form).
- There is **a balance between computational costs (latency) and quality of results**. The loose combination of project images showing potential useful sources of information, and text, even when text is not fully accurate, mitigates the impact of hallucinations and makes these less obvious - perhaps minimising their impact in practical terms - to a casual user.

- The implementation of many of the **"advanced" features has proven to be less impactful than expected** and, on this use case, not worth the extra latency.

5.3 Ethical, social, legal and professional considerations

This research has been conducted in accordance with the ethics procedures of Northumbria University. All data used to feed the RAG has been sourced from the public domain, and no information about the test users has been collected beyond anonymous answers to the survey. No legal nor social concerns have arisen.

From a security and professional standpoint, a **local LLM has been used on purpose to mimic the conditions that might happen on a real-life project**, where a company would want to protect their intellectual property by keeping their data on-premises, instead of connecting to a possibly more advanced or larger LLM through a third-party API, with the risk of it accessing their private data.

5.4 Personal reflections

This project has been an opportunity to learn, develop and test a fully working Python-based solution composed of many moving parts, for the purpose of answering the research question. There has been a **steep but productive learning process involving the many different languages and frameworks used**, and the fundamental concepts behind them. The final prototype shows a proof of concept for a real-world system that could easily be repackaged for production as a service within a wider web application - or just stay as a standalone application in itself.

5.5 Recommendations

Future research on RAG for large architectural practices might want to consider some of the below recommendations from this experience:

- **The LLM and vector database components have proven to be, by far, the most impactful components of a RAG** throughout this study. Even with the relatively small-sized LLM used (llama3.2), most questions returned a reasonable answer within the top-k results (Hit@K = 1).

- For the AD industry, even a basic RAG would be capable of delivering "good enough" results for high level, domain-specific search (especially if supported by an image grid) - even if such results were not absolutely perfect, the enhanced semantic search capability inherent to **RAG offers a faster and more intuitive way to find potentially relevant information** than, for example, Windows Explorer.
- Some of the major enhancement additions to the basic RAG architecture (reranker, rewriter, captions) seemed to need their own dedicated side research and optimisation to really have a significant impact.
- Likewise, there was an expectation for the **visual question answering** feature to improve the responses. However, **the latency increase and poor semantic understanding of the images** had this feature removed from the final prototype.
- While this study has addressed the most essential data format (PDF containing images and text), other relevant formats, **very specific to the AD context have been left out and might be of great interest for future studies**, such as *BIM* and *CAD* formats, which typically hold spatial and construction information about projects.
- **Test user survey has proven insightful and engaging**. Inviting potential end users of these products to give feedback can provide valuable qualitative insights, not possible to achieve with just quantitative methods.

5.6 Conclusion

This study has tried, by making the best use of limited available hardware and resources, to determine whether RAG is a viable approach to KM in the AD industry. As opposed to other text-heavier industries, like legal or finance, the visual-centric approach of architectural design offers a more forgiving environment for the application of this technology, where users might still be able to find -or get a hint of - what they are looking for on a response accompanied by an image grid, even if the the text answer is not fully accurate.

Results and feedback received suggest it might be a **promising paradigm** for users at corporations to access know-how, **combining the power of AI through the use of natural language, while keeping the security and privacy that is required to protect intellectual property**.

REFERENCES

Ng, K.K.Y., Matsuba, I. and Zhang, P.C. (2025) 'RAG in Health Care: A Novel Framework for Improving Communication and Decision-Making by Addressing LLM Limitations.' *NEJM AI*, 2(1). Available at: https://doi.org/10.1056/aira2400380

Wedel, K. (2025) 'The Insight Layer: Defining a New Category for Context-Aware Organizational Intelligence Through AI-Augmented Insight Delivery'. Available at: https://osf.io/njgtk/download (Accessed: 08/05/25)

Taherdoost, H. (2022) 'What Are Different Research approaches? Comprehensive Review of qualitative, quantitative, and Mixed Method research, Their applications, types, and Limitations'. *Journal of Management Science & Engineering Research*, 5(1), pp.53–63. Available at: https://doi.org/10.30564/jmser.v5i1.4538

Tanaka, R. et al. (2025) 'VDocRAG: Retrieval-Augmented Generation over Visually-Rich Documents'. Available at: https://doi.org/10.48550/arXiv.2504.09795

Kahn, A. et al. (2024) 'Developing Retrieval Augmented Generation (RAG) based LLM Systems from PDF: An Experience Report'. Available at: https://doi.org/10.48550/arXiv.2410.15944

Lewis, P. et al. (2020) Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, *NeurIPS 2020*. Available at: https://doi.org/10.48550/arXiv.2005.11401

Sirota, A. (2025) *Introduction to RAG solution from scratch - Building a RAG Solution from Scratch*. LinkedIn. Available at: https://www.linkedin.com/learning/building-a-rag-solution-from-scratch/intro-to-rag-solution-from-scratch?u=74773554 (Accessed 9 May 2025)

pixegami (2024) *Python RAG Tutorial (with Local LLMs): AI For Your PDFs*. YouTube. Available at: https://www.youtube.com/watch?v=2TJxpyO3ei4 (Accessed 9 May 2025)

Pello, R. (2019) *Design science research — a summary*. Medium. Available at: https://medium.com/@pello/design-science-research-a-summary-bb538a40f669. (Accessed: 11/05/25)

Hevner, A. et al. (2004) *Design Science in Information Systems Research*. Available at: http://doi.org/10.2307/25148625

Yu, H. et al. (2024) *Evaluation of Retrieval-Augmented Generation: A Survey*. Available at: https://doi.org/10.48550/arXiv.2405.07437

Creswell, J.W. (2014) *Research design: Qualitative, quantitative & mixed methods approaches*. 4th ed. Los Angeles: Sage.

Choi, B., Poon, S.K. and Davis, J.G. (2008) 'Effects of knowledge management strategy on organizational performance: A complementarity theory-based approach', *Omega (Oxford)*, 36(2), pp. 235–251. Available at: https://doi.org/10.1016/j.omega.2006.06.007

Nakash, M. and Bouhnik, D. (2022) 'Can return on investment in knowledge management initiatives in organizations be measured?', *ASLIB JOURNAL OF INFORMATION MANAGEMENT*, 74(3), pp. 417–431. Available at: https://doi.org/10.1108/AJIM-09-2021-0268

Dalkir, K. and Liebowitz, J. (2011) *Knowledge management in theory and practice*. 2nd ed. Cambridge, Mass.: MIT Press.

Koenig, M. (2012) *What is KM? Knowledge Management Explained*. Available at: https://www.kmworld.com/Articles/Editorial/What-is-KM-Knowledge-Management-Explained-82405.aspx. (Accessed: 01/05/25)

Davenport, T.H. and Bean, R. (2025) *The GenAI Focus Shifts to Innovation at Colgate- Palmolive*. Available at: https://sloanreview.mit.edu/article/the-genai-focus-shifts-to-innovation-at-colgate-palmolive/ (Accessed: 14/05/25)

Şahin, G., Varol, K. and Pak, B.K. (2024) 'LLM and RAG-Based Question Answering Assistant for Enterprise Knowledge Management' *2024 9th International Conference on Computer Science and Engineering (UBMK)* (pp. 1-6). IEEE. Available at: https://doi.org/10.1109/UBMK63289.2024.10773564

Jarrahi, M.H. et al. (2023) 'Artificial intelligence and knowledge management: A partnership between human and AI' *Business Horizons*, 66(1), pp.87-99. Available at: https://doi.org/10.1016/j.bushor.2022.03.002

Fan, W. et al. (2024) 'A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models'. *arXiv preprint*. Available at: https://doi.org/10.48550/arxiv.2405.06211

Fan, W. et al. (2025) 'Towards Retrieval-Augmented Large Language Models: Data Management and System Design', *2025 IEEE 41st International Conference on Data Engineering (ICDE), Hong Kong, 2025*, pp. 4509-4512, Available at: https://doi.org/10.1109/ICDE65448.2025.00341

Chorev, S. (2024) *Top RAG Metrics for Enhanced Performance*. Available at: https://www.deepchecks.com/top-rag-metrics-for-enhanced-performance/ (Accessed: 04/06/25).

TruEra (2024) *What is the RAG Triad?*. Available at: https://truera.com/ai-quality-education/generative-ai-rags/what-is-the-rag-triad/ (Accessed: 04/06/25)

Hassan, M. (2025) *Data Collection – Methods Types and Examples.* Available at: https://researchmethodology.org/data-collection/ (Accessed: 05/06/25)

Ni, B. et al. (2025) Towards Trustworthy Retrieval Augmented Generation for Large Language Models: A Survey. Available at: https://doi.org/10.48550/arXiv.2502.06872

Peacock, A. (2025) *Foster + Partners unveils 100,000-seat stadium for Manchester United*. Available at: https://www.dezeen.com/2025/03/11/manchester-united-stadium-foster-partners/ (Accessed: 05/06/25)

Bruckhaus, T. (2024) 'RAG Does Not Work for Enterprises'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2406.04369

Chen, J. et al. (2025) ‘From Standalone LLMs to Integrated Intelligence: A Survey of Compound AI Systems’. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2506.04565

huggingface.co. (n.d.) *Advanced RAG on HuggingFace documentation using LangChain*. Available at: https://huggingface.co/learn/cookbook/en/advanced_rag (Accessed: 18/08/25)

Joshi, P. et al. (2024) ‘Robust Multi Model RAG Pipeline For Documents Containing Text, Table & Images’. *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC),* Salem, India, 2024, pp. 993-999. Available at: https://doi.org/10.1109/icaaic60222.2024.10574972

purpleSlate (2024). *The Ultimate Guide to Understanding Advanced Retrieval Augmented Generation Methodologies*. Available at: https://medium.com/@social_65128/the-ultimate-guide-to-understanding-advanced-retrieval-augmented-generation-methodologies-467cd05a2ecd (Accessed 18 Aug. 2025).

Yuan, X. et al. (2025) 'mKG-RAG: Multimodal Knowledge Graph-Enhanced RAG for Visual Question Answering'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2508.05318

Bonomo, M. and Bianco, S. (2025) 'Visual RAG: Expanding MLLM visual knowledge without fine-tuning'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2501.10834

Chan, C. et al. (2024) 'RQ-RAG: Learning to Refine Queries for Retrieval Augmented Generation'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2404.00610

Kuang, J. et al. (2025) 'Natural Language Understanding and Inference with MLLM in Visual Question Answering: A Survey', *ACM Computing Surveys*, 57(8), pp. 1-36. Available at: https://doi.org/10.48550/arXiv.2411.17558

Xia, S., Cheng, Y. and Tian, R. (2024) ‘ARCHICLIP. Enhanced Contrastive Language–Image Pre-training Model With Architectural Prior Knowledge’. *CAADRIA 2024*. Available at: https://doi.org/10.52842/conf.caadria.2024.1.069

Noh, H. et al. (2017) 'Large-Scale Image Retrieval with Attentive Deep Local Features'. *ICCV 2017*, pp. 3456-3465. Available at: https://doi.org/10.48550/arXiv.1612.06321

Li, S. et al. (2025) 'Enhancing Retrieval-Augmented Generation: A Study of Best Practices'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2501.07391

Li, J., Yuan, Y. and Zhang, Z. (2024) 'Enhancing LLM Factual Accuracy with RAG to Counter Hallucinations: A Case Study on Domain-Specific Queries in Private Knowledge-Bases'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2403.10446

Dong, G. et al. (2025) ‘Understand What LLM Needs: Dual Preference Alignment for Retrieval-Augmented Generation’. *Proceedings of the ACM on Web Conference 2025*. Available at: https://doi.org/10.48550/arXiv.2406.18676

Mahapatra, J. and Garain, U. (2024) 'Impact of Model Size on Fine-tuned LLM Performance in Data-to-Text Generation: A State-of-the-Art Investigation'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2407.14088

Zhao, S. et al. (2025) 'Understanding the Design Decisions of Retrieval-Augmented Generation Systems'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2411.19463

Brown, T. B. et al. (2020) 'Language Models are Few-Shot Learners'. *NeurIPS 2020*, Vancouver. Available at: https://doi.org/10.48550/arXiv.2005.14165

Agrawal, A., et al. (2015) 'VQA: Visual Question Answering'. *ICCV 2015*. Available at: https://doi.org/10.48550/arXiv.1505.00468

Huynh, N. D., et al. (2025) 'Visual Question Answering: from early developments to recent advances - a survey'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2501.03939

Radford, A., et al. (2021) 'Learning Transferable Visual Models From Natural Language Supervision'. *Proceedings of the 38th International Conference on Machine Learning*, pp. 8748-8763. Available at: https://doi.org/10.48550/arXiv.2103.00020

Li, J., et al. (2022) 'BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation'. *International Conference on Machine Learning, PMLR*. Available at: https://doi.org/10.48550/arXiv.2201.12086

Li, J., et al. (2023) 'BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models'. *International Conference on Machine Learning, PMLR*. Available at: https://doi.org/10.48550/arXiv.2301.12597

Alayrac, J. B., et al. (2022) 'Flamingo: a Visual Language Model for Few-Shot Learning'. *NeurIPS*. Available at: https://doi.org/10.48550/arXiv.2204.14198

Liu, H., et al. (2023) 'Visual Instruction Tuning'. *NeurIPS*. Available at: https://doi.org/10.48550/arXiv.2304.08485

Kaplan, J., et al. (2020) 'Scaling Laws for Neural Language Models'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2001.08361

Mikolov, T., et al (2013) 'Distributed Representations of Words and Phrases and their Compositionality'. *NeurIPS*. Available at: https://doi.org/10.48550/arXiv.1310.4546

Norouzi, M., et al., (2013) 'Zero-Shot Learning by Convex Combination of Semantic Embeddings'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.1312.5650

Kiros, R., et al. (2014) 'Unifying Visual-Semantic Embeddings with Multimodal Neural Language Models'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.1411.2539

Taipalus, T. (2024) 'Vector database management systems: Fundamental concepts, use-cases, and current challenges', *Cognitive Systems Research*, 85, pp. 101216. Available at: https://doi.org/10.48550/arXiv.2309.11322

Zhuang, S. et al. (2023) 'Bridging the Gap Between Indexing and Retrieval for Differentiable Search Index with Query Generation'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2206.10128

Tay, Y. et al. (2022) 'Transformer Memory as a Differentiable Search Index'. *NeurIPS 2022*. Available at: https://doi.org/10.48550/arXiv.2202.06991

Dai, Z. et al., (2022) 'Promptagator: Few-shot dense retrieval from 8 examples'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2209.11755

Ma, X. et al. (2023) 'Query Rewriting for Retrieval-Augmented Large Language Models'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2305.14283

Devine, P. (2025) 'ALoFTRAG: Automatic Local Fine Tuning for Retrieval Augmented Generation'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2501.11929

Lee, Z. P., Lin, A. and Tan, C. (2025) 'Finetune-RAG: Fine-Tuning Language Models to Resist Hallucination in Retrieval-Augmented Generation'. Available at: https://doi.org/10.48550/arXiv.2505.10792

Barnett, S. et al. (2024) 'Seven Failure Points When Engineering a Retrieval Augmented Generation System'. *Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering*. Available at: https://doi.org/10.48550/arXiv.2401.05856

Xue, J. et al. (2024) 'BadRAG: Identifying Vulnerabilities in Retrieval Augmented Generation of Large Language Models'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2406.00083

Jain, A. et al. (2017) 'Information Retrieval using Cosine and Jaccard Similarity Measures in Vector Space Model', *International Journal of Computer Applications*, 164(6). Available at: https://doi.org/10.5120/ijca2017913699

Baek, J., Aji, A. F. and Saffari, A. (2023) 'Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2306.04136

Sharma, C. (2025) 'Retrieval-Augmented Generation: A Comprehensive Survey of Architectures, Enhancements, and Robustness Frontiers'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2506.00054

Zhong, K. et al. (2025) 'SemRAG: Semantic Knowledge-Augmented RAG for Improved Question-Answering'. *arXiv preprint*. Available at: https://doi.org/10.48550/arXiv.2507.21110

Kumar, B.R.V. et al. (2025) 'Prospects of Retrieval-Augmented Generation (RAG) for Academic Library Search and Retrieval', *Information technology and libraries*, Vol.44 (2), pp. pp .1-15. Available at: https://doi.org/10.13140/RG.2.2.33848.25607

Pinecone (no date) *Rerankers and Two-Stage Retrieval*. Available at: https://www.pinecone.io/learn/series/rag/rerankers/ (Accessed: 20/09/25)

NVIDIA (2025) *Build an Enterprise RAG pipeline*. Available at: https://build.nvidia.com/nvidia/build-an-enterprise-rag-pipeline (Accessed: 20/09/25)

Şakar, T. and Emekci, H. (2025) 'Maximizing RAG efficiency: A comparative analysis of RAG methods', *Natural Language Processing*, 31(1), pp. 1-25. Available at: https://doi.org/10.1017/nlp.2024.53

Asai, A. et al. (2024) 'Self-RAG: Learning to retrieve, generate, and critique through self-reflection', *ICLR 2024*. Available at: https://doi.org/10.48550/arXiv.2310.11511

Friel, R., Belyi, M. and Sanyal, A. (2025) 'RAGBench: Explainable Benchmark for Retrieval-Augmented Generation Systems'. Available at: https://doi.org/10.48550/arXiv.2407.11005

IBM (no date) *Retrieval Augmented Generation*. Available at: https://www.ibm.com/architectures/hybrid/genai-rag (Accessed: 25/10/25).

Cohen, D. et al. (2025) *WixQA*. Available at: https://huggingface.co/datasets/Wix/WixQA (Accessed: 28/10/25)

Brooke, J. (1995) 'SUS - A quick and dirty usability scale'. *Usability Evaluation in Industry* (pp.189-194). Available at: https://www.researchgate.net/publication/319394819_SUS_--_a_quick_and_dirty_usability_scale (Accessed: 02/11/2025)

Knowledge Architecture (no date) *Synthesis*. Available at: https://www.knowledge-architecture.com/ (Accessed: 05/11/2025)

Xiahou, X. et al. (2025) 'Knowledge Management in Construction Quality Management: Current State, Challenges, and Future Directions', *IEEE Transactions on Engineering Management.* Available at: https://doi.org/10.1109/TEM.2025.3550354

Olayinka, R. et al. (2015) 'Impact of knowledge management on the cost of poor quality', *Management Procurement and Law*, 168(4), pp. 177-188. Available at: https://doi.org/10.1680/mpal.1400035

Yap, J., Lim, B. and Skitmore, M. (2022) 'Capitalising knowledge management (KM) for improving project delivery in construction'., *Ain Shams Engineering Journal, 13(6).* Available at: https://doi.org/10.1016/j.asej.2022.101790

Klesel, M., Whittmann, H. (2025) 'Retrieval-Augmented Generation (RAG)', *Bus Inf Syst Eng 67, 551–561.* Available at: https://doi.org/10.1007/s12599-025-00945-3

Baur, D. et al. (2025) 'Development and Evaluation of a Retrieval-Augmented Generation Chatbot for Orthopedic and Trauma Surgery Patient Education: Mixed-Methods Study'. *JMIR AI 2025 4*. Available at: https://doi.org/10.2196/75262

Python Software Foundation (2025) *Python 3.11.14*. Available at: https://www.python.org/downloads/release/python-31114/ (Accessed: 14/11/25)

Richardson, L. (2025) *Beautiful Soup*. Available at: https://pypi.org/project/beautifulsoup4/ (Accessed: 14/11/25)

ReportLab (2025) *ReportLab*. Available at: https://www.reportlab.com/ (Accessed: 14/11/25)

Dezeen (2025) Available at: https://www.dezeen.com/ (Accessed: 15/11/25)

Peacock, A. (2025) *Foster + Partners unveils 100,000-seat stadium for Manchester United.* Available at: https://www.dezeen.com/2025/03/11/manchester-united-stadium-foster-partners/ (Accessed: 15/11/25)

Mozilla Developer Network (2025) *JavaScript*. Available at: https://developer.mozilla.org/en-US/docs/Web/JavaScript (Accessed: 15/11/25)

ECMA International (2025) *ECMAScript*. Available at: https://ecma-international.org/publications-and-standards/standards/ecma-262/ (Accessed: 15/11/25)

W3C (2025) *HTML Living Standard*. Available at: https://html.spec.whatwg.org/multipage/ (Accessed: 15/11/25)

W3C (2023) *CSS Display Module Level 3*. Available at: https://www.w3.org/TR/css-display-3/ (Accessed: 15/11/25)

Mueller, A. (2025) *WordCloud*. Available at: https://amueller.github.io/word_cloud/ (Accessed: 15/11/25)

Matplotlib.org (2025) *Matplotlib*. Available at: https://matplotlib.org/ (Accessed: 15/11/25)

huggingface.co (2025) *SentenceTransformers*. https://huggingface.co/sentence-transformers (Accessed: 15/11/25)

scikit-learn.org (2025) *Scikit-Learn*. Available at: https://scikit-learn.org/stable/ (Accessed: 15/11/25)

Pallets Projects (2025) *Flask*. Available at: https://flask.palletsprojects.com/en/stable/ (Accessed: 15/11/25)

Tailwind CSS (2025) *Tailwind CSS*. Available at: https://tailwindcss.com/ (Accessed: 15/11/25)

Optuna.org (2025) *Optuna*. Available at: https://optuna.org/ (Accessed: 15/11/25)

Microsoft (2025) *Microsoft Forms*. Available at: https://forms.office.com/ (Accessed: 15/11/25)

Chroma (2025) *Chroma*. Available at: https://www.trychroma.com/ (Accessed: 15/11/25)

LangChain (2025) *LangChain*. Available at: https://www.langchain.com/ (Accessed: 15/11/25)

Ollana (2025) *Ollama*. Available at: https://ollama.com/ (Accessed: 15/11/25)

Pillow (2025) *Pillow*. Available at: https://pillow.readthedocs.io/en/stable/ (Accessed: 15/11/25)

Pytorch (2025) *Pytorch*. Available at: https://pytorch.org/ (Accessed: 15/11/25)

huggingface.co (2025) *Transformers*. Available at: https://huggingface.co/docs/transformers/en/index (Accessed: 15/11/25)

huggingface.co (2025) *Sentence Transformers*. Available at: https://huggingface.co/sentence-transformers (Accessed: 15/11/25)

SQLAlchemy (2025) *SQLAlchemy. Available at:* https://www.sqlalchemy.org/ (Accessed: 15/11/25)

Ollama (2025) *Ollama*. Available at: https://ollama.com/download (Accessed: 16/11/25)

Prettier (2025) *Prettier.* Available at: https://prettier.io/ (Accessed: 17/11/25)

APPENDIX

Including:

Code structure diagram
Sample PDF report
Baseline parameter values
Test user survey form questions
Test user survey results

Code structure diagram

PE7046.ini (Codebase structure diagram showing only most relevant files)

```
PE7046
├── .env
├── .gitignore
├── app
│   ├── database.py
│   ├── extensions.py
│   ├── routes
│   │   ├── components.py
│   │   ├── main.py
│   │   ├──
│   │   └── __init__.py
│   ├── static
│   │   ├── js
│   │   │   ├── functions.js
│   │   │   └── main.js
│   │   └── projects
│   │       └── ...
│   ├── templates
│   │   ├── base.html
│   │   ├── footer.html
│   │   ├── header.html
│   │   ├── loading.html
│   │   ├── project_generic.html
│   │   ├── project_thumbs.html
│   │   └── rag.html
│   ├── __init__.py
│   └── __main__.py
├── config.py
├── environment-cpu.yml
├── environment.yml
├── instance
│   ├── production.db
│   └── test.db
├── Makefile
├── pytest.ini
├── README.md
├── requirements-cpu.txt
├── requirements.txt
├── run.py
├── services
│   └── rag
│       ├── caption_generator.py
│       ├── client.py
│       ├── db
│       │   ├── ...
│       │   ├── chroma.sqlite3
│       │   └── txt
│       │       └── chunks.txt
│       ├── generator.py
│       ├── query.py
│       ├── rag_db.py
```

```
│       ├── reranker.py
│       ├── retriever.py
│       ├── settings.py
│       ├── settings_heavy.py
│       ├── vqa.py
│       ├── __init__.py
│       ├── __main__.py
│       └── __main__.py
├── tests
│   ├── test_caption_generator.py
│   ├── test_client.py
│   ├── test_generator.py
│   ├── test_query.py
│   ├── test_rag_db
│   │   ├── ...
│   │   ├── data
│   │   │   └── projects
│   │   │       └── ...
│   │   ├── img
│   │   │   └── ...
│   │   └── txt
│   │       └── chunks_test.txt
│   ├── test_rag_db.py
│   ├── test_reranker.py
│   ├── test_retriever.py
│   ├── test_vqa.py
│   └── __init__.py
└── utility
    ├── data_analysis
    │   ├── da_plotter.py
    │   ├── output
    │   │   ├── heatmap.png
    │   │   ├── heatmap_additional.png
    │   │   ├── wordcloud.png
    │   │   └── wordcount_bar.png
    │   ├── semantic.py
    │   ├── text
    │   │   ├── full_text_output.txt
    │   │   └── individual
    │   │       └── ...
    │   ├── text_reader.py
    │   ├── words.py
    │   ├── __init__.py
    │   ├── __main__.py
    │   └── __main__.py
    ├── db_setup
    │   ├── data
    │   │   └── ...
    │   ├── pdf_extractor.py
    │   ├── seed.py
    │   ├── __init__.py
    │   └── __main__.py
    ├── misc
    │   └── util.py
```

```
    ├── rag_evaluation
    │   ├── benchmarker.py
    │   ├── bench_rag_db
    │   │   ├── ...
    │   │   └── txt
    │   │       └── chunks_bench.txt
    │   ├── csv_*
    │   │   └── ...
    │   ├── evaluation_summary.ipynb
    │   ├── evaluator.py
    │   ├── metrics.py
    │   ├── plots
    │   │   ├── first_pass_plot.png
    │   │   └── second_pass_plot.png
    │   ├── profiler.py
    │   ├── rag_config.py
    │   ├── __init__.py
    │   └── __main__.py
    └── scraper
        ├── pdf_generator.py
        ├── scraper.py
        ├── url_list.py
        ├── __init__.py
        └── __main__.py
```

Sample PDF report (Scraped article from dezeen.com)

Foster + Partners unveils 100,000-seat stadium for Manchester United

British studio Foster + Partners has revealed designs for a stadium covered by "a vast umbrella" for English football club Manchester United, which will become the biggest stadium in the UK. Forming part of the regeneration of the Old Trafford area in Manchester, the 100,000-seat stadium is set to be built next to the football team's current 74,197-capacity Old Trafford stadium. Foster + Partners has unveiled its design for a 100,000-seat stadium in ManchesterOnce complete, the Foster + Partners-designed stadium will surpass the UK's current largest football stadium, Wembley Stadium, which has a capacity of 90,000. Renders show the stadium covered in Manchester United's signature red colour, enveloped by a tent-like structure that extends over public outdoor space. A tent-like structure will cover the stadium"The stadium is contained by a vast umbrella, harvesting energy and rainwater and sheltering a new public plaza that is twice the size of Trafalgar Square," said Foster + Partners founder Norman Foster. "The outward-looking stadium will be the beating heart of a new sustainable district, which is completely walkable, served by public transport, and endowed by nature," he continued. "It is a mixed-use miniature city of the future – driving a new wave of growth and creating a global destination that Mancunians can be proud of." It will become the biggest stadium in the UKFoster + Partners' design is to be built next to Manchester United's Old Trafford stadium, which has been the team's home ground for 115 years. It will aim to enhance the experience for visiting football fans with improved views of the pitch and acoustics. "This has to be one of the most exciting projects in the world today, with incredible regional and national significance," said Foster. "It all starts with the fans' experience, bringing them closer than ever to the pitch and acoustically cultivating a huge roar." Public space will surround the venueReferencing Manchester United's trident emblem, the tent-like structure will be propped up by three masts. Foster + Partners plans to complete the stadium in five years by using prefabricated elements transported via the Manchester Ship Canal – a waterway linking Manchester to the Irish Sea. It will form part of the regeneration of the Old Trafford area"Three towering masts hold up the stadium's canopy, embodying the club's trident emblem and creating a distinctive presence on the skyline," Foster + Partners head of studio Nigel Dancey told Dezeen. "Teamwork and the city's renowned spirit of innovation are cornerstones of this remarkable project, which would look to embrace the network of the Manchester Ship Canal and cutting-edge prefabrication technologies to build the stadium in record time." The design aims to be "the world's greatest football stadium"Foster + Partners' stadium will aim to have facilities that make it one of the best football venues in the world while paying homage to the ground's history. The wider regeneration of Old Trafford intends to deliver 92,000 jobs and 17,000 homes to the area. "Today marks the start of an incredibly exciting journey to the delivery of what will be the world's greatest football stadium, at the centre of a regenerated Old Trafford," said Manchester United co-owner Jim Ratcliffe. "Our current stadium has served us brilliantly for the past 115 years, but it has fallen behind the best arenas in world sport," Ratcliffe added. "By building next to the existing site, we will be able to preserve the essence of Old Trafford while creating a truly state-of-the-art stadium that transforms the fan experience only footsteps from our historic home." Other stadiums recently unveiled around the world include a sunken football pitch in Egypt by Gensler and Populus's plans to revamp the Giuseppe Sinigaglia Stadium at Lake Como. The images are courtesy of Foster + Partners.

Baseline parameter values

```
# BASELINE PARAMETERS
# rag type
BASELINE_RAG_TYPE = TEXT

# LLM model parameters
BASELINE_LLM = "llama3.2:latest"
BASELINE_TEXT_EMBEDDING_MODEL = "nomic-embed-text"

# general retrieval and weighting settings
BASELINE_INITIAL_K_SAMPLES = 18 # preliminary K_Samples before reranking
BASELINE_K_SAMPLES = 9 # set to 9 for production to max out interface grid
BASELINE_TEXT_WEIGHT = 1.0
BASELINE_IMG_WEIGHT = 1.0
BASELINE_CAPS_WEIGHT = 1.0

# generation settings
BASELINE_BATCH_SIZE = 4
BASELINE_MAX_NEW_TOKENS = 60
BASELINE_MIN_NEW_TOKENS = 20
BASELINE_DECODING = "greedy"
BASELINE_NUM_BEAMS = 1
BASELINE_NO_REPEAT_NGRAM_SIZE = 3
BASELINE_LENGTH_PENALTY = 1.0

# performance
BASELINE_NUM_IO_WORKERS = 8
BASELINE_CACHE_ANSWERS = True

# captions flag
BASELINE_CAPTION_FLAG = True

# reranker flag & settings
BASELINE_USE_RERANKER = False
BASELINE_RERANKER_MODEL = "cross-encoder/ms-marco-MiniLM-L-6-v2"
BASELINE_RERANKER_BATCH_SIZE = 32

# rewriter flag, settings and domain hints
BASELINE_USE_REWRITER = False
BASELINE_REWRITER_MODEL = "zorowin123/rq_rag_llama2_7B"
BASELINE_DOMAIN_HINTS = ("architecture, architectural design, buildings,"
                         "building elements, building structures, public spaces, "
                         "facades, interior spaces.")
```

Test user survey form questions

RAG System Assessment

You are being invited to take part in a research study. My name is Salvador Cristobal Navarro Perez and I am a student of MSc Computer Science with Artificial Intelligence at Northumbria University. Please take a few moments to read this introductory information before continuing on to the survey.

1. Introductory Statement:
Researcher's Name: Salvador Cristobal Navarro Perez
Name of Researcher's School: Computer and Information Sciences
Title of Research: "Retrieval-augmented generation for knowledge management at large architecture practices"

2. What is this research about?
This project explores the effectiveness and utility of a retrieval-augmented generation (RAG) system forknowledge management.

3. Why am I doing this research?
To assess if an AI-powered knowledge management system for architecture could be useful and saveresources at the workplace.

4. Why have you been invited to take part?
Because you are a professional working in the architectural design industry.

5. How will your data be used?
To help determine if the RAG is useful. The results will be presented in summary form for academic purposes so no individual can be identified.

6. What will happen if you decide to take part in this research study?
This survey is completely voluntary and anonymous. If you decide to take part in this research, the data you contribute by completing the surveys may contribute to the aims of this research. You may skip any questions.

7. How will your privacy be protected?
If you decide to take part in this research, you will be assigned a unique identifier code which will replace your name in the data. Thus all data will be anonymous. Nobody will have information to identify you. All data will be processed in agreement with GDPR 2018 and the Data Protection Act2018.

8. What are the benefits of taking part in this research study?
Understanding of knowledge management strategies in the architecture industry will hopefully benefit from you taking part in this research.

9. What are the risks of taking part in this research study?
The risks of taking part in this research are minimised. You will only be asked questions relative to the usability of the prototype.

10. Can you change your mind and withdraw at any point?
You can change your mind and withdraw at any point during the survey before you click submit, your data will not be used in any way with no penalty. However, as the data is being collected anonymously, there will be no way to withdraw your data once you have clicked submit.

11. How will you find out about what happens with this project?
By contacting the lead researcher, Salvador Cristobal Navarro Perez. Some of the results might be published, and if they are you will be notified of this. Data will be anonymous and no participant will be identifiable.

Contact details for further information:
Lead Researcher: Salvador Cristobal Navarro Perez – salvador.perez@northumbria.ac.uk

Consent Form

1. Do you understand and accept the above terms? *

- ◯ Yes
- ◯ No

RAG-Specific Questions

Open the application on your browser, get a bit familiar with the projects and ask some questions that have the answers somewhere in the project texts.

Answers will be more accurate if some of the information exists in the project description text.

Try a few questions which have answers in the text of one of the projects of your choice and questions which do not have them.

Imagine the system could include links to key documents of the projects in the future, and bear this in mind when you assess how useful this could be.

2. How relevant were the text answers returned? *

☆ ☆ ☆ ☆ ☆

3. Did the text answers feel factually accurate? *

☆ ☆ ☆ ☆ ☆

4. Were the project images returned relevant to your question? *

☆ ☆ ☆ ☆ ☆

5. Did the combination of text and project images help? *

☆ ☆ ☆ ☆ ☆

6. When you asked some question where the information was not present on the text description, how accurate was the response? *

☆ ☆ ☆ ☆ ☆

7. Was the answering speed acceptable? *

☆ ☆ ☆ ☆ ☆

8. Do you think this tool could become potentially useful as a way to find key information about projects? *

☆ ☆ ☆ ☆ ☆

9. How useful do you think this tool could be (provided projects have a reasonably good text description), compared to, i.e. just manually finding files with Windows Explorer? *

☆ ☆ ☆ ☆ ☆

10. What additional feature(s) or changes do you think would make this tool more useful than it is right now? *

11. Do you imagine any other potentially useful applications of this tool, or an evolved version of it? (i.e. marketing, visual communication, etc.) *

This content is neither created nor endorsed by Microsoft. The data you submit will be sent to the form owner.

Microsoft Forms

Test user survey results

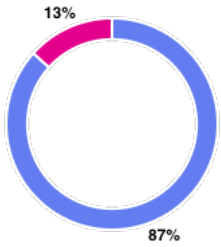
Responses Overview

Active

Responses	Average Score	Average Time
15	0	10:45

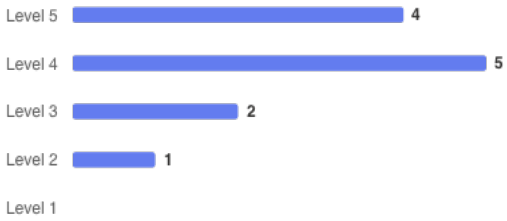
1. Do you understand and accept the above terms? (0 point)

Yes	13
No	2



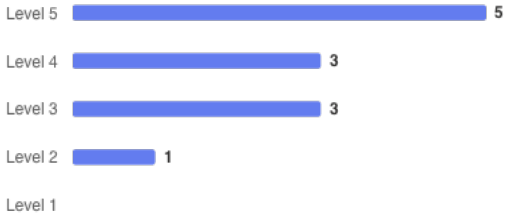
2. How relevant were the text answers returned? (0 point)

4.00
Average Rating



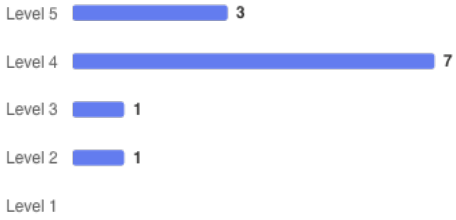
3. Did the text answers feel factually accurate? (0 point)

4.00
Average Rating



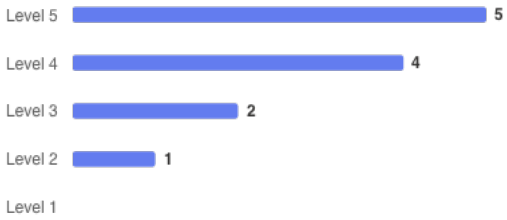
4. Were the project images returned relevant to your question? (0 point)

4.00
Average Rating



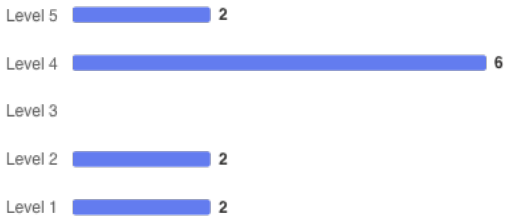
5. Did the combination of text and project images help? (0 point)

4.08
Average Rating



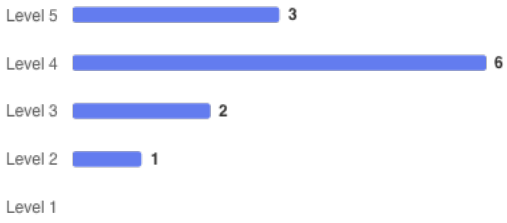
6. When you asked some question where the information was not present on the text description, how accurate was the response? (0 point)

3.33
Average Rating



7. Was the answering speed acceptable? (0 point)

3.92
Average Rating

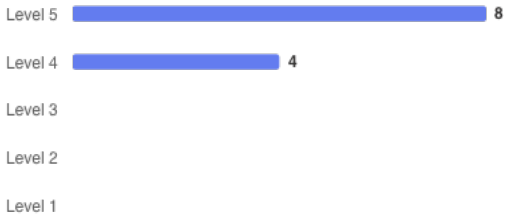


8. Do you think this tool could become potentially useful as a way to find key information about projects? (0 point)

4.75
Average Rating

9. How useful do you think this tool could be (provided projects have a reasonably good text description), compared to, i.e. just ma nually finding files with Windows Explorer? (0 point)

4.67
Average Rating



10. What additional feature(s) or changes do you think would make this tool more useful than it is right now? (0 point)

Northumbria University
NEWCASTLE

Salvador Cristobal Navarro Perez
w23051337